

AI Development Capacity & Design Generation Engine

Codename ENVELOPE — Version 2.0

Codename: ENVELOPE **Supersedes:** PRD V1 (retained for reference; its worked examples are withdrawn as arithmetically invalid) **Version:** 2.0 **Status:** For build **Change basis:** Red-team review. Product vision unchanged; delivery restructured into six phases; fifteen specific corrections applied and tracked in §29.3.

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Reading guide

Priorities: **P0** blocks the phase · **P1** required for phase completion, degradable · **P2** next phase · **P3** later.

Provenance classes used throughout: **DERIVED** (from a cited rule) · **ASSUMED** (declared default) · **USER_SET** · **OBSERVED** (from precedent, Phase 2+) · **TRADEOFF** (optimizer choice, Phase 5).

Two labels appear frequently and mean exactly what they say: - **OPEN REGULATORY QUESTION** — we do not know the answer and have not invented one. The system asks the user and records their answer as **USER_SET**. - **HEURISTIC** — a number produced by a method we have not calibrated. Never displayed without the label.

1. Product Vision

1.1 The vision is unchanged

An **AI Development Capacity Engine** that answers, for a specific plot:

Given this plot, its regulations, the developer's standards, the project brief, and relevant approved precedents — what is the best development configuration that can legally and commercially be built here?

The long-term pipeline remains exactly as conceived:

PLOT → REGULATIONS → DEVELOPER STANDARDS → RFP → HISTORICAL PRECEDENTS
→ GENERATE → OPTIMIZE → VALIDATE → ARCHITECTURAL OUTPUTS
→ EXPLAIN

The product is **not** a compliance checker. Validation is the mechanism that bounds generation, not the deliverable. The product works **before** architectural design exists.

1.2 What changed in V2

Nothing about the destination. Everything about the order of arrival.

V1 attempted to build the pipeline as one system. V2 builds it as six layers, each of which is independently testable, independently sellable, and each of which **cannot be safely built before the layer beneath it is proven**. The reason is not caution for its own sake — it is that every upper layer's correctness is defined in terms of the lower layer's outputs. You cannot evaluate whether a generated massing is good until you can compute, exactly and defensibly, what volume it is allowed to occupy. You cannot calibrate a precedent-based estimate until you have a deterministic estimate to calibrate against. You cannot review an AI-extracted rule until you have a rule model to review it into.

1.3 The layering, stated as dependency rather than sequence

Phase 5	OPTIMIZATION	needs a generator whose outputs are trustworthy
▲		
Phase 4	GENERATION	needs constraints that are both checkable and constructive
▲		
Phase 3	AI EXTRACTION	needs a rule model to extract INTO, and a review workflow
▲		
Phase 2	PRECEDENT	needs a deterministic baseline to calibrate AGAINST
▲		
Phase 1	CONFIGURATIONS	needs a capacity engine to configure WITHIN
▲		
Phase 0	CAPACITY ENGINE	needs nothing but geometry, rules, and arithmetic

Each arrow is a hard technical dependency, not a preference. This is the entire argument of V2.

2. Product Strategy

2.1 Strategic position

	Statement
Core asset	A versioned, cited, human-approved, machine-executable model of Dubai development regulation, plus a deterministic geometry engine that can <i>construct</i> from it, not merely <i>check</i> against it.
Wedge	Defensible capacity in minutes, with every number traceable.
Expansion	Configurations → precedent intelligence → document automation → generation → optimization.
Moat, ascending durability	Typology library < precedent corpus < executable rule base < approval-outcome feedback loop.
Non-moat	The LLM, the UI, the optimizer. All commodity.

2.2 The three claims the product must never conflate

This is a product-defining distinction and it appears in every output:

Claim	What it means	Can we support it?
Self-consistency	The configuration satisfies the constraints we encoded, and its arithmetic closes	Yes, from Phase 0
Rule coverage	The constraints we encoded are the ones that apply	Partially; quantified and disclosed
Professional agreement	A qualified architect would produce a comparable answer	Measured, not asserted (\$22)
Regulatory validity	The authority would approve it	Never claimed. Not obtainable.

Non-negotiable Principle 7: the validator agreeing with the generator is *self-consistency*, nothing more. No output, no report, no marketing material may describe it as compliance.

2.3 Phase gating

Each phase ships to real users before the next begins. Each phase has explicit kill/re-scope criteria (\$26). A phase that fails its criteria does not proceed on hope.

Phase	Name	Duration	Ships to
0	Proof of Engine	8-12 weeks	1-2 design partners, free
1	Development Configuration Engine	10-14 weeks	3-5 paying customers
2	Precedent Intelligence	12-16 weeks	Same, as upgrade
3	AI Document Understanding	10-12 weeks	Same, reduces onboarding cost
4	AI Generation	16-20 weeks	Gated on architect panel
5	Full Optimization	12-16 weeks	Gated on Phase 4

Cumulative to full vision: **~18-24 months**. This is longer than V1's 11-month estimate and the difference is honesty about the knowledge-authoring workstream and about the sequencing dependencies above.

2.4 Non-negotiable product principles

These carry forward from the brief, unchanged, and are enforced structurally rather than by policy. Where enforcement is mechanical, the mechanism is named.

#	Principle	Enforcement
1	Historical projects are precedent evidence, not regulatory authority	Type-level: <code>PrecedentObservation</code> cannot appear in a <code>ConstraintSet</code> derivation path. CI assertion.
2	Every important output traces to a rule, an assumption, an objective, or a precedent	Provenance class is non-nullable on every emitted value. CI assertion.
3	Every emitted configuration passes an invariant/conservation layer	§12. Fails the build and blocks emission.
4	The invariant layer is independent of the generator	Separate module, separate owner, no shared imports. Import linter.
5	Hard constraints separated from soft preferences	Distinct types; hard constraints are feasibility filters, never penalty terms.
6	Five distinct capacity concepts, never merged	§15. Each has its own field and its own derivation.
7	Never claim compliance because the validator agrees with the generator	§2.2. Report template makes the distinction structural.
8	Distinguish self-consistency, rule coverage, geometric validity, regulatory validity, professional agreement	§16.5 — reported as five separate statements.
9	Every rule carries source, citation, effective date, jurisdiction, applicability, version, approval, provenance	Schema: all NOT NULL. §11.
10	Shared assets are versioned and approval-controlled	§21. Rules, defaults, area definitions, typologies, normalizations, deferred classifications.

3. PHASE 0 — Proof of Engine

3.1 Goal

Prove that a deterministic engine can take a real Dubai residential-tower plot and produce a buildable envelope and capacity calculation that falls **within the disagreement band of two practising architects**, in seconds, with every number traceable.

Note the accuracy target is *relative to human disagreement*, not to an imaginary true value. If two qualified architects differ by 6% on the same plot and the engine lands between them, the engine is as good as the profession — and it is faster, reproducible, and auditable, which is the product.

3.2 Scope

In: - One Dubai community, one land-use type, residential towers - Form-based plot input; manual boundary drawing on a basemap - Rectilinear and simple convex polygon plots only - ~40 mechanized rules as typed records - Buildable envelope: per-edge setbacks, height, FAR, coverage, tower plate cap - Parking demand → parking area → level count → podium implication - Three capacity bands: **regulation-limited, geometry-limited, parking-limited** - User-entered realism discount (a `USER_SET` multiplier, clearly not a system estimate) - Assumption register with perturbation sensitivity - Full provenance graph, click-through - Invariant / conservation layer - Independent validation module - JSON + PDF output

Out: LLM anything · optimization · multiple configurations · floor plans · precedent · similarity · jurisdiction overlays · DXF · realistic capacity band.

3.3 Why no LLM in Phase 0

Not because AI is deprecated in this product. Because:

1. **Extraction is not the risky part.** Document extraction is a known, bounded engineering problem with commodity solutions. Capacity computation with defensible provenance is not.
2. **AI extraction has nothing to extract into until the rule model exists.** Phase 3 depends on Phase 0's schema. Building extraction first means building it twice.
3. **A form takes 90 seconds.** For 10 plots in one community, manual entry costs less than the extraction pipeline costs to debug.
4. **It isolates the variable.** If Phase 0 fails, we need to know it failed on geometry and rules, not on OCR.

Phase 3 adds AI extraction on top of a proven rule model, with a review workflow that already has somewhere to send candidates.

3.4 Phase 0 output — what a user receives

1. Buildable envelope (podium footprint, tower plate cap, height ceiling), with **the binding constraint named for each dimension**
2. Three capacity bands with derivations
3. Parking demand, area, level count, and headroom
4. Applicable rules with citations, plus rules considered and excluded with reasons
5. Assumption register, sensitivity-ranked
6. Invariant check results
7. Validation results, with the five-way claim distinction (§2.2)

8. Deferred-check list — what was applicable but not assessed

9. PDF report + JSON

3.5 Phase 0 success criteria

#	Criterion	Threshold
P0-S1	Inter-architect variance measured and published	Measured (any value)
P0-S2	Engine result within the two-architect band	8 of 10 plots
P0-S3	Every emitted value has a provenance class and complete derivation	100%
P0-S4	Invariant layer passes on all emitted configurations	100%
P0-S5	Runtime, full computation	< 10 s
P0-S6	Design-partner users open the provenance tree unprompted	≥ 2 of 3 sessions
P0-S7	Mechanization rate and authoring throughput measured	Measured and reported
P0-S8	Land-pack completeness rate measured	Measured and reported

P0-S1, S7 and S8 are **measurement** criteria, not performance criteria. Phase 0's most valuable output is not the software — it is four numbers that size the rest of the business (§26).

4. PHASE 1 — Development Configuration Engine

4.1 Goal

Move the output from "what can fit?" to "what are the reasonable ways to develop this?"

4.2 Additions

- **Developer Standards** — structured library per organization, hard/soft classification elicited at onboarding, reconciliation against rules
- **RFP / Brief** — structured targets, contradiction detection before computation
- **Parameterized tower typologies** — a small architect-authored library (4-6), each with a bounded parameter vector
- **Configuration enumeration** — a coarse deterministic grid over typology parameters, not a search algorithm
- **Objective functions** — explicit terms, explicit normalization, explicit weights
- **Configuration comparison** — aligned metrics, decomposable scoring
- **Developer-specific priorities** — organization objective policy with governance (§20.4)
- **Commercially preferred capacity** — the fourth capacity concept (§15)

4.3 Deliberately still absent

No NSGA-II. No CP-SAT. No Pareto front. Phase 1 enumerates a grid of perhaps 50-500 configurations, evaluates them all deterministically, and ranks them. This is not an optimizer; it is exhaustive evaluation of a coarse space. It is enough to answer the customer question, and it defers the entire performance problem (§8.4) until there is evidence the search space is genuinely too large to enumerate.

If exhaustive enumeration turns out to be sufficient at production granularity, **Phase 5 may be substantially reduced or skipped**. This is a real possible outcome and would be good news.

4.4 Phase 1 success criteria

#	Criterion	Threshold
P1-S1	Distinct configurations produced per project	3-6, distinct in <i>design</i> space (§18.3)
P1-S2	Every configuration passes invariants and validation	100%
P1-S3	Brief contradictions detected before computation	100% of arithmetic contradictions
P1-S4	Ranking is fully decomposable to term contributions	100%
P1-S5	Enumeration completes	< 60 s
P1-S6	Paying customers	≥ 3
P1-S7	Recommended configuration carried into design without material capacity revision	≥ 40% (baseline measurement)

5. PHASE 2 — Precedent Intelligence

5.1 Goal

Introduce approved-project data as **evidence**, and only then introduce the historically-observed capacity concept.

5.2 Two acquisition paths, planned in parallel

Neither is assumed to succeed. The phase is scoped so that **either one alone is sufficient** to deliver reduced value, and failure of both degrades the product gracefully rather than blocking it.

Path A — Public / municipal data

- Published municipal open-data channels (building permits, building summary information, floor-level information, building usages, project records) under a commercial licence permitting derivative analytical use
- **Strength:** volume, breadth, low relationship cost
- **Weakness:** almost certainly lacks NSA, unit mix, and *permitted* FAR — which are precisely the fields the calibration needs. Path A alone supports density, floor-count, and parking-ratio benchmarking, **not** efficiency or achieved-vs-permitted FAR.

Path B — Developer-partner portfolio

- 10–40 completed projects per partner, contributed under contract, with full internal records
- **Strength:** contains NSA, mix, permitted FAR, efficiency — the fields that matter
- **Weakness:** small n, selection bias toward one developer's practice, contractual complexity

The honest consequence

The metrics that make a "historically observed capacity" credible require **Path B**. Path A cannot produce them. This is stated here rather than discovered in month nine. Phase 2 planning assumes Path B is the primary source and Path A is breadth support.

5.3 Deliverables

- Historical project schema with data tiering and licence tagging (§17.1)
- Similarity engine — faceted, regime-gated, per-domain transferability (§17.4)
- Precedent retrieval and pattern extraction with minimum-n enforcement
- Benchmarking: where does this configuration sit against comparable approvals
- Precedent-based calibration → **the historically-observed capacity concept**

5.4 The realistic capacity band — entry condition

The band that V1 shipped in MVP is **removed from Phase 0 and Phase 1 entirely**. It may be introduced only when all of the following hold:

Condition	Threshold
Observations in the conditioning cell	$n \geq 8$
Fields required present	plot area, permitted FAR, achieved GFA, NSA, unit count

Condition	Threshold
Regulatory regime match	Same regime, or divergence explicitly flagged
Backtest coverage on held-out projects	$\geq 80\%$ within P10–P90
Backtest sharpness	$\text{median} (P90 - P10) / P50 \leq 20\%$

Coverage without sharpness is meaningless — a band of "3,000 to 30,000 m²" achieves 100% coverage and zero value. Both are required.

Below $n = 8$, the system returns `INSUFFICIENT_PRECEDENT` and the user keeps their `USER_SET` realism discount from Phase 0.

5.5 Phase 2 success criteria

#	Criterion	Threshold
P2-S1	Precedent records at Tier B completeness	≥ 120 (Path B ≥ 40)
P2-S2	Top-5 retrieved precedents judged relevant by an architect	$\geq 70\%$
P2-S3	Backtest coverage and sharpness both met	Per §5.4
P2-S4	Zero precedent values in any constraint derivation path	100%, CI-asserted
P2-S5	Every precedent record carries source, licence, redistributable flag	100%

6. PHASE 3 — AI Document Understanding

6.1 Goal

Reduce the cost of getting data into the system, without letting a model become an authority.

6.2 Scope

AI extraction for: regulation documents, developer standards, RFPs. Plus requirement classification, assumption identification, conflict identification, and the human review workflow that governs all of it.

6.3 The mandatory pipeline — no exceptions

```
Source document
→ LLM extraction to a strict candidate schema
→ DETERMINISTIC NORMALIZATION
  · unit conversion · parameter-vocabulary binding · type coercion
  · arithmetic cross-checks · duplicate detection
→ CANDIDATE RULE (status = DRAFT, never evaluable)
→ Human review by a qualified reviewer
→ Approval (second reviewer for life-safety and planning parameters)
→ Versioned rule with validity window
→ Production
```

An LLM output can never reach production as an authoritative rule. Enforced at the schema level: `rule.approved_by` is NOT NULL and the rule evaluator refuses to load records with `status != APPROVED`. There is no configuration flag that bypasses this.

6.4 Deterministic-first extraction

Developer standards documents are 60–80% tables. The pipeline routes structured content to a deterministic table parser and sends only prose to the LLM. Unit normalization, area cross-validation, and parameter binding are separate deterministic passes over model output — never instructions given to the model.

6.5 Extraction confidence — the labelling rule

Self-consistency across samples measures sampling variance, not correctness. Therefore:

Label	Condition	Permitted use
CALIBRATED_CONFIDENCE	Validated against the gold set, reliability curve published, Brier score reported	May gate the review queue
HEURISTIC_CONFIDENCE	Anything else	Display only, always labelled, may not gate anything

A field is never presented with a bare number. It is `CALIBRATED_CONFIDENCE 0.94` or `HEURISTIC_CONFIDENCE 0.94 (uncalibrated)`.

6.6 Phase 3 success criteria

#	Criterion	Threshold
P3-S1	Critical-field extraction accuracy on gold set	$\geq 95\%$
P3-S2	Rules reaching production without human approval	0
P3-S3	Statements silently dropped (neither extracted nor queued)	0
P3-S4	Confidence labelled calibrated or heuristic	100%
P3-S5	Onboarding time reduction vs. manual	$\geq 50\%$

7. PHASE 4 — AI Generation

7.1 Goal

Generate massing, core, floorplate, units, and typical floors — within constraints, never outside them.

7.2 Scope

Tower massing generation · core configuration · floorplate generation · unit planning · typical floor generation · multiple design alternatives.

7.3 The generation contract

The LLM does not invent geometry. Geometry is produced by deterministic parametric instantiation and solver-based packing. If any generative model is used for layout proposal in future, its output is a **proposal** that must pass constraint enforcement, invariant checks, and independent validation before it can be emitted — the same gate any other candidate passes.

7.4 Fidelity ceiling

Generated: core component footprints, corridor centreline and width, unit zone polygons with type, area, frontage, depth, orientation, adjacency.

Not generated: internal partitions, rooms, wall thickness, doors, windows, structural grid, balcony geometry beyond area allowance.

Stated in the drawing, not just the report.

7.5 Geometry restriction lift

Phase 0 restricts to rectilinear and simple convex plots. Phase 4 is where non-convex and irregular footprints are addressed, because unit packing on arbitrary polygons is the genuinely hard case (§14.5) and should be attempted only with the rest of the stack proven.

7.6 Phase 4 gate — pre-authorized deferral

#	Criterion	Threshold
P4-S1	Generated floors geometrically valid	100%
P4-S2	Invariant layer pass	100%
P4-S3	Blind architect panel: "reasonable starting point a designer could develop"	≥ 70%
P4-S4	Inter-rater reliability on the panel	Krippendorff $\alpha \geq 0.6$

If P4-S3 lands below 60%, Phase 4 does not ship. Phases 0–3 continue as the product and Phase 4 returns to development. This deferral is authorized here, in advance, so it is not a schedule argument later.

8. PHASE 5 — Full Optimization

8.1 Entry condition

Phase 5 begins only when Phase 4's generator is reliable — meaning P4-S1 through S4 are met and stable across two consecutive evaluation rounds. Optimizing an untrustworthy generator produces confidently wrong answers faster.

8.2 Scope

Multi-objective optimization · Pareto frontier · weight sensitivity · advanced search · CP-SAT where justified · surrogate models · caching · performance engineering.

8.3 "Where justified"

CP-SAT and NSGA-II are introduced **only if** Phase 1's exhaustive enumeration has been demonstrated insufficient at production granularity. The evidence required is a measured configuration space that cannot be enumerated within the runtime budget at the granularity customers need. Absent that evidence, enumeration stays.

8.4 The performance problem, costed honestly

If a nested architecture is adopted (evolutionary outer loop over a solver-based inner loop), the arithmetic must be done before the design is chosen:

Population 100×50 generations = 5,000 evaluations. At 2 s per inner packing solve, that is ~2.8 CPU-hours; on 32 cores, ~5 minutes wall clock — but only if packing is genuinely 2 s and only if the outer loop parallelizes cleanly against a solver with its own threading. A realistic first implementation lands at **30-90 minutes**.

Mitigations that must be designed in, not discovered: surrogate models for the inner loop, caching keyed on floorplate-geometry hash, and pre-solving packing as a lookup over (footprint $\times$ core position) so the inner solve leaves the hot path entirely. This is real design work and it is scoped into Phase 5, not assumed away.

8.5 Phase 5 success criteria

#	Criterion	Threshold
P5-S1	Hard-constraint violations in returned set	0
P5-S2	Returned options distinct in design space	Per §18.3 metric
P5-S3	Full run wall clock	≤ 10 min p95
P5-S4	Re-rank on weight change without re-search	≤ 2 s
P5-S5	Immutable-run reproducibility (§13.4)	100%

9. Functional Requirements

Format: **ID · Requirement · Rationale · Input · Processing · Output · Priority · Acceptance Criteria · Dependencies.**

Prefixes: **DEF** definitions · **PLT** plot & geometry · **RUL** rule model · **CAP** capacity · **PRK** parking · **INV** invariants · **VAL** validation · **PRV** provenance · **ASM** assumptions · **OUT** output · **STD** standards · **RFP** brief · **CFG** configuration · **OBJ** objectives · **PRE** precedent · **SIM** similarity · **EXT** extraction · **GEN** generation · **OPT** optimization · **GOV** governance.

9.1 Phase 0 — Foundations

FR-DEF-001 — Area & Metric Definitions Annex

- **Requirement:** A single versioned, human-approved document defining every area and ratio term the system uses, cited by every report.
- **Rationale:** GFA, BUA, NSA, sellable area, efficiency and FAR are used inconsistently across the industry. Without a signed definition, every customer dispute becomes unresolvable and every metric is untestable. This is the root dependency of the entire capacity model.
- **Input:** Regulatory definitions where they exist; practitioner convention; architect and customer review.
- **Processing:** Draft → architect review → design-partner review → sign-off → version.
- **Output:** **MetricDefinition** records (§10.2); a one-page annex referenced by ID in every report.
- **Priority:** P0 — **week 1, before any code**
- **Acceptance Criteria:**
 - AC1: Defines at minimum: GFA, BUA, NSA, Sellable Area, Gross Efficiency, Tower Efficiency, Parking Area, Bay Area Factor, FAR, Plot Coverage.
 - AC2: Each definition states explicit inclusions and exclusions (balconies, terraces, plant, cores, circulation, parking, basements).
 - AC3: Records whether parking counts toward FAR, or marks it **OPEN REGULATORY QUESTION** (see FR-DEF-002).
 - AC4: Versioned; every report cites the version used.
 - AC5: No code merges that computes an area term absent from the annex.
- **Dependencies:** None. Everything else depends on it.

FR-DEF-002 — Parking-in-FAR Determination

- **Requirement:** The system must explicitly represent whether parking area counts toward FAR/GFA, and must not assume an answer.
- **Rationale:** This single question changes capacity by 15-35% on a podium-parking tower. Guessing it silently is the highest-magnitude quiet error the system can make.
- **Input:** Regulatory research; authority confirmation; user declaration.
- **Processing:** If an authoritative answer is confirmed and citable → a **Rule**, class **DERIVED**. If not → status **OPEN_REGULATORY_QUESTION**, and the user is required to declare a treatment at project setup, recorded as **USER_SET**.
- **Output:** **parking_counts_toward_far** with a provenance class; both scenarios computable on demand.
- **Priority:** P0
- **Acceptance Criteria:**

- AC1: The value is never `ASSUMED`. It is `DERIVED` (cited) or `USER_SET` (declared). There is no default.
- AC2: While `OPEN_REGULATORY_QUESTION`, the project cannot compute capacity until the user declares a treatment.
- AC3: The report states the treatment applied, on the face page, with its provenance class.
- AC4: A one-click comparison shows capacity under both treatments.
- **Dependencies:** FR-DEF-001.

FR-PLT-001 — Plot Input & Boundary

- **Requirement:** Capture plot identity, planning parameters and boundary geometry via form entry and manual polygon drawing.
- **Rationale:** Phase 0 isolates the engine. A form removes extraction as a variable and costs the user ~90 seconds.
- **Input:** User form; basemap; optional coordinate list.
- **Processing:** Validate polygon simplicity, closure, non-self-intersection; compute area, principal axes, minimum bounding rectangle, convexity ratio, per-edge length and bearing; user classifies each edge (`ROAD` with hierarchy / `ADJACENT_PLOT` / `OPEN_SPACE` / `OTHER`); cross-check computed area against stated area.
- **Output:** `Plot` + `PlotEdge[]` with geometry provenance `USER_SET`.
- **Priority:** P0
- **Acceptance Criteria:**
 - AC1: Only rectilinear and simple convex polygons accepted in Phase 0; others rejected with a clear message naming the restriction.
 - AC2: Computed vs. stated area deviation > 2% blocks until resolved.
 - AC3: Edge classification is mandatory for every edge; no default.
 - AC4: A trained user completes plot entry in ≤ 3 minutes, measured.
- **Dependencies:** FR-DEF-001.

Design note. Edge classification is deliberately *not* automated in Phase 0. Reliable automatic road-edge detection needs a road-hierarchy GIS layer joined to cadastral edges, whose availability and join quality are unverified (§30). Making the user classify four edges is a 20-second cost that removes an unproven dependency from the critical path.

FR-RUL-001 — Typed Rule Records & Evaluator Registry

- **Requirement:** Rules are stored as typed data records evaluated by a small registry of named evaluator functions. **No DSL, no compiler in Phase 0.**
- **Rationale:** A DSL designed before ~100 rules have been authored will encode the wrong abstractions and will be rewritten. Typed records deliver bitemporality, provenance, testability and versioning immediately, while leaving awkward rules expressible as registered Python evaluators. Extract a DSL later, when the shape is known — it will be a better DSL.
- **Input:** Source instrument; architect authoring.
- **Processing:** Author record → bind to parameter vocabulary → attach citation → author ≥ 3 tests → review → approve → publish with validity window.
- **Output:** `Rule` records (§11.2).
- **Priority:** P0
- **Acceptance Criteria:**
 - AC1: Every rule carries source, citation, effective date, jurisdiction, applicability, version, approval status, provenance — all non-null (Principle 9).

- AC2: `approved_by` is non-null and is a qualified human. The evaluator refuses to load unapproved records. No bypass flag exists.
- AC3: Every rule has ≥ 3 passing tests including one boundary and one negative case.
- AC4: Every rule declares its class: `GENERATIVE`, `FILTERING`, or `EVALUATIVE_ONLY` (§11.3).
- AC5: Bitemporal — the rule set as of any past date is reconstructible.
- AC6: Evaluator registry has ≤ 12 functions at Phase 0. Exceeding this is a signal the abstraction is wrong; review before adding.
- **Dependencies:** FR-DEF-001.

FR-RUL-002 — Applicability & Rule Set Materialization

- **Requirement:** Select applicable rules for a project and freeze them as an immutable set bound to the run.
- **Rationale:** Reproducibility and defensibility both require knowing exactly which rules were applied.
- **Input:** Project context (land use, height band, plot attributes, occupancy); rule base at a version.
- **Processing:** Evaluate applicability predicates; resolve overlaps by the resolution model (§11.5); materialize snapshot.
- **Output:** `ApplicableRuleSet` with per-rule inclusion reason and near-miss exclusion reasons.
- **Priority:** P0
- **Acceptance Criteria:**
 - AC1: Immutable once bound. Re-running after a rule-base change creates a new run.
 - AC2: User can see why each rule was included and why notable rules were excluded.
 - AC3: Non-scalar conflicts are handled per §11.6 and never silently resolved.
- **Dependencies:** FR-RUL-001.

FR-RUL-003 — Rule Class Compilation

- **Requirement:** Compile applicable rules into three distinct operational forms.
- **Rationale:** V1 required every rule to be "constructive." That is impossible for topological rules — there is no constructive inverse of a travel-distance limit. Forcing one abstraction produced a spec that could not be implemented.
- **Input:** `ApplicableRuleSet`.
- **Processing:** Partition into `GENERATIVE` (construct geometry or bounds directly), `FILTERING` (prune candidates during construction), `EVALUATIVE_ONLY` (assess after construction; may only reject), and `DEFERRED` (applicable, not mechanizable at this fidelity).
- **Output:** `GenerativeSet`, `FilterSet`, `EvaluativeSet`, `DeferredCheckList`.
- **Priority:** P0
- **Acceptance Criteria:**
 - AC1: Every applicable rule lands in exactly one partition. Unassigned rules fail the build.
 - AC2: `DeferredCheckList` appears in every output, in the body.
 - AC3: `EVALUATIVE_ONLY` rules can only reject a candidate, never modify one.
- **Dependencies:** FR-RUL-002.

FR-PLT-002 — Buildable Envelope Solver

- **Requirement:** Compute the buildable envelope from plot geometry and generative rules, naming the binding constraint for each dimension.
- **Rationale:** The atomic unit of value. Everything above it is built on this being exactly right.
- **Input:** `Plot`, `PlotEdge[]`, `GenerativeSet`.
- **Processing:** Apply per-edge setback offsets $\rightarrow$ intersect $\rightarrow$ setback-permitted footprint. Apply coverage cap. Podium footprint = $\min(\text{setback-permitted}, \text{coverage cap})$. Apply tower plate cap.

Apply height ceiling and floor-to-floor to derive maximum level counts. Recompute every area by an independent method and compare.

- **Output:** `BuildableEnvelope` : setback-permitted footprint, coverage cap area, podium footprint, tower plate cap, height ceiling, max level counts — **each with its binding constraint and rule citation**.
- **Priority:** P0
- **Acceptance Criteria:**
 - AC1: Geometry valid, simple, closed; independent area recomputation agrees to $\leq 0.1\%$.
 - AC2: Every dimension names its binding constraint and cites its rule.
 - AC3: Where two constraints bind within 1%, both are reported.
 - AC4: Runs in < 2 s for ≤ 12 edges.
 - AC5: Golden set (§23): engine result within the two-architect band on ≥ 8 of 10 plots.
- **Dependencies:** FR-PLT-001, FR-RUL-003.

FR-PRK-001 — Parking Demand, Area and Level Implication

- **Requirement:** Compute parking demand, required area, achievable level count, and the resulting capacity implication.
- **Rationale:** Parking frequently determines podium size, which determines the ground plane and tower footprint. Computing GFA first and parking second produces schemes that collapse on contact with reality.
- **Input:** Unit schedule (or candidate unit count and mix), parking rules, bay area factor, available basement and podium levels.
- **Processing:** Apply per-type ratios $\rightarrow$ resident bays; apply visitor and accessible provisions; total bays. Bays $\times$ bay-area-factor $\rightarrow$ required area. Available area = $\Sigma(\text{level footprint} \times \text{usable fraction})$. Derive supply in bays, headroom, and the maximum unit count supportable.
- **Output:** `ParkingResult` : bays by category with citations, required area, supply, headroom, supportable unit ceiling.
- **Priority:** P0
- **Acceptance Criteria:**
 - AC1: Bay counts derive from cited rules, per category.
 - AC2: Bay area factor is explicit, editable, and stated in the report — it is a top-three sensitivity driver and must never be hidden.
 - AC3: Feeds capacity band C before any ranking or recommendation.
 - AC4: **No bay-level layout is produced.** Output is a quantum, an area, and a level count.
 - AC5: Headroom reported in bays and percent.
- **Dependencies:** FR-PLT-002, FR-RUL-003.

FR-CAP-001 — Capacity Bands A / B / C

- **Requirement:** Produce three separately-derived capacity figures.
- **Rationale:** A single capacity number conceals which constraint is actually binding, which is the most decision-relevant fact on the page.
- **Input:** `BuildableEnvelope`, `ParkingResult`, applicable rules, area definitions.
- **Processing:** **A — Regulation-limited:** FAR ceiling and any explicit area caps. **B — Geometry-limited:** what the envelope physically holds under height and footprint limits, ignoring FAR. **C — Parking-limited:** the largest scheme the achievable parking supply supports. Report the minimum as governing, and name it.
- **Output:** `CapacityBands` with per-band derivation, binding constraint, and headroom to the next-binding constraint.
- **Priority:** P0

- **Acceptance Criteria:**
- AC1: All three bands always shown. Showing only one is a P0 defect.
- AC2: The governing band is named explicitly.
- AC3: Integer-granularity losses (e.g. unused FAR because another floor does not fit) are reported, not absorbed.
- AC4: **No "realistic" or "expected" band exists in Phase 0.** Absence is enforced by schema.
- **Dependencies:** FR-PLT-002, FR-PRK-001, FR-DEF-001.

FR-CAP-002 — User-Entered Realism Discount

- **Requirement:** Allow the user to apply a discount factor to the governing band, recorded as `USER_SET`.
- **Rationale:** Practitioners know their own realization gap. The system does not, until Phase 2. Letting the user supply it is honest; inventing it is not.
- **Input:** User-entered factor, default 1.00.
- **Processing:** Apply to governing band; propagate to dependent metrics; record in the assumption register.
- **Output:** Discounted capacity, class `USER_SET`, with the user's identity attached.
- **Priority:** P1
- **Acceptance Criteria:**
- AC1: Default is 1.00 (no discount). The system never pre-fills a discount.
- AC2: Labelled `USER_SET` with the user's name wherever displayed.
- AC3: Report states plainly that this is a user judgement, not a system estimate.
- **Dependencies:** FR-CAP-001.

FR-INV-001 — Invariant / Conservation Layer

- **Requirement:** An independent layer asserting algebraic identities on every emitted configuration, before emission.
- **Rationale:** V1's own worked example failed by factors of ~3 because nothing checked that areas summed. Constraints check *permissibility*; invariants check *arithmetic coherence*. They are different, and only the second would have caught it.
- **Input:** Any `Configuration` or `CapacityResult`.
- **Processing:** Evaluate the invariant catalogue (§12.2). Any failure blocks emission and raises an internal defect.
- **Output:** `InvariantReport`: per-check pass/fail with computed vs. expected and residual.
- **Priority:** P0
- **Acceptance Criteria:**
- AC1: Runs on every emitted artifact including test fixtures, worked examples, and documentation samples.
- AC2: A failure **blocks emission**. It is never a warning.
- AC3: Implemented in a module with no imports from the generator or capacity engine (Principle 4), enforced by an import linter in CI.
- AC4: Owned by an engineer who does not own the capacity engine.
- AC5: Every example in every product document passes it, verified in CI.
- **Dependencies:** FR-DEF-001.

FR-VAL-001 — Independent Validation

- **Requirement:** Validate configurations against hard constraints, soft preferences and evaluative-only rules, in a module independent of generation.

- **Rationale:** Shared code between generator and validator makes the validator confirm the generator's own bugs.
- **Input:** Configuration, `FilterSet`, `EvaluativeSet`, `PreferenceSet`.
- **Processing:** Evaluate each; record per-check result with citation.
- **Output:** `ValidationReport`.
- **Priority:** P0
- **Acceptance Criteria:**
 - AC1: No shared evaluation code with the generator; separate module, separate owner.
 - AC2: Every result cites its rule.
 - AC3: **The report states explicitly that agreement between validator and generator is self-consistency, not compliance** (Principle 7).
 - AC4: The five-way claim distinction (§16.5) appears verbatim.
- **Dependencies:** FR-RUL-003, FR-INV-001.

FR-PRV-001 — Provenance Graph

- **Requirement:** Every emitted value carries a provenance class and a complete derivation path.
- **Rationale:** Traceability is the product. It is also unretrofitable — provenance must be emitted at every layer boundary, so adding it later means rewriting every layer.
- **Input:** All engine steps.
- **Processing:** Emit nodes and edges at each derivation; assign classes; link to rule citations, assumptions, user inputs.
- **Output:** Provenance DAG per run.
- **Priority:** P0
- **Acceptance Criteria:**
 - AC1: 100% of emitted values have a non-null provenance class. CI-asserted; a null class fails the build.
 - AC2: Every `DERIVED` value's path terminates in a cited rule.
 - AC3: Derivation tree renders in ≤ 300 ms.
 - AC4: No `PrecedentObservation` appears in any constraint's derivation path (Principle 1). CI-asserted from Phase 2.
- **Dependencies:** All Phase 0 engines.

FR-ASM-001 — Assumption Register with Sensitivity

- **Requirement:** Surface every `ASSUMED` value, ranked by measured impact, editable inline, acknowledged before export.
- **Rationale:** The core failure mode is a user mistaking an assumption for a fact. This screen is the mitigation and the trust-establishing moment.
- **Input:** Provenance graph.
- **Processing:** Collect `ASSUMED` values; perturb each by $\pm 10\%$ (or ± 1 unit for integers); recompute; rank by effect on governing capacity.
- **Output:** Ranked register with value, basis, and quantified impact.
- **Priority:** P0
- **Acceptance Criteria:**
 - AC1: Non-skippable before first export.
 - AC2: Impact quantified, e.g. "+3 m²/bay removes 1 floor: −1,060 m² GFA, −8 units."
 - AC3: Every entry editable inline with immediate recompute.
 - AC4: Rendered in a visually distinct state (amber) everywhere the value appears, not only in the register.
- **Dependencies:** FR-PRV-001.

FR-OUT-001 — Report and JSON Export

- **Requirement:** PDF report and complete JSON export.
 - **Rationale:** The report is the artifact that goes to the investment committee; the JSON is what makes the product composable.
 - **Input:** Run outputs.
 - **Processing:** Template composition; JSON serialization of the full model including provenance.
 - **Output:** PDF + JSON.
 - **Priority:** P0
 - **Acceptance Criteria:**
 - AC1: Face page carries: metric-definitions version, rule-set version, parking-in-FAR treatment and its provenance, engine version, reviewer, professional-use disclaimer.
 - AC2: Every number footnoted to its provenance.
 - AC3: Body contains the assumption register, the deferred-check list, the invariant report and the five-way claim statement.
 - AC4: JSON round-trips: re-importing reproduces the identical report.
 - **Dependencies:** All Phase 0.
-

9.2 Phase 1 — Configurations

FR-STD-001 — Developer Standards Library

- **Requirement:** Per-organization standards library with hard/soft classification and reconciliation against rules.
- **Rationale:** Whether a standard is contractually mandatory or merely preferred is the largest source of downstream error, and standards documents almost never say. Inferring it will be wrong roughly a third of the time, invisibly.
- **Input:** Standards documents; a guided elicitation session at onboarding.
- **Processing:** Structure into parameter-bound requirements; **elicit hard/soft classification from the customer's design manager once per document**; reconcile against the rule set into `COMPATIBLE` / `MORE_RESTRICTIVE` / `LESS_RESTRICTIVE` / `ORTHOGONAL`.
- **Output:** `ReconciledStandardSet` with dispositions and effective values.
- **Priority:** P0 (Phase 1)
- **Acceptance Criteria:**
 - AC1: Classification is elicited, never inferred, in Phase 1.
 - AC2: Every `LESS_RESTRICTIVE` standard produces a visible record: regulation governs, with quantified impact.
 - AC3: Library is versioned and reused across that organization's projects.
 - AC4: A standard cannot override a rule except through a documented variance (FR-GOV-003).
- **Dependencies:** FR-RUL-002.

FR-RFP-001 — Brief Structuring and Contradiction Detection

- **Requirement:** Structure brief targets and detect internal and regulatory contradictions **before** computation.
- **Rationale:** Catching "180 units needs 21,900 m² against a 16,000 m² ceiling" in 30 seconds prevents an error that today survives weeks into a project.
- **Input:** Structured form (Phase 1) or document (Phase 3).
- **Processing:** Type each target `HARD_TARGET` / `SOFT_TARGET` / `ASPIRATION`; run the contradiction battery (§16.2); compare against band A.

- **Output:** `ProjectBrief` with typed targets, `Contradictions`, `BriefGaps`.
- **Priority:** P0 (Phase 1)
- **Acceptance Criteria:**
 - AC1: All arithmetic contradictions detected before computation, naming the incompatible pair and quantifying the shortfall.
 - AC2: Targets exceeding band A are flagged with the governing rule cited.
 - AC3: Gaps become assumption-register entries; they are never silently defaulted.
- **Dependencies:** FR-CAP-001.

FR-CFG-001 — Typology Library and Configuration Enumeration

- **Requirement:** Enumerate configurations across an architect-authored parametric typology library.
- **Rationale:** Free-form generation produces novel, unbuildable, unexplainable geometry and destroys traceability. Curated typologies give most of the value with full explainability.
- **Input:** `BuildableEnvelope`, typology library, standards, brief.
- **Processing:** For each typology, enumerate a coarse grid over its bounded parameter vector; reject candidates failing filter rules at instantiation; evaluate the survivors exhaustively.
- **Output:** `Configuration[]` with parameter vectors and metrics.
- **Priority:** P0 (Phase 1)
- **Acceptance Criteria:**
 - AC1: ≥ 4 typologies attempted where the envelope permits; infeasible typologies report why.
 - AC2: 100% of emitted configurations pass invariants and independent validation.
 - AC3: Typology library is data-driven; adding a typology requires no code deployment.
 - AC4: Enumeration completes in < 60 s.
 - AC5: **No search algorithm.** Exhaustive evaluation only.
- **Dependencies:** FR-PLT-002, FR-INV-001, FR-VAL-001.

FR-OBJ-001 — Objective Functions and Ranking

- **Requirement:** Score and rank configurations under an explicit, versioned objective profile.
- **Rationale:** Weights are an organizational value judgement. A ranking without a recorded profile is neither reproducible nor defensible.
- **Input:** Configurations; `ObjectiveProfile`.
- **Processing:** Normalize each term via a **versioned, approval-controlled normalization function** (§21); apply weights; produce decomposable scores; compute margin over runner-up; test robustness under weight perturbation.
- **Output:** Ranked configurations, scorecards, recommendation, robustness statement.
- **Priority:** P0 (Phase 1)
- **Acceptance Criteria:**
 - AC1: Score fully decomposable to term contributions.
 - AC2: If the top two are within 3%, the system states the recommendation is **not robust** and explains what distinguishes them.
 - AC3: Every report states the objective profile ID and version.
 - AC4: Re-ranking on weight change does not re-enumerate; completes in < 2 s.
 - AC5: **No fictional objectives.** Marketability and view quality are excluded until the underlying data exists.
- **Dependencies:** FR-CFG-001, FR-GOV-002.

FR-CAP-003 — Commercially Preferred Capacity

- **Requirement:** Report the capacity of the configuration the developer's own objectives select, as a distinct concept from the three regulatory/physical bands.
- **Rationale:** Principle 6 requires five distinct capacity concepts. This is the fourth.
- **Input:** Ranked configurations; objective profile.
- **Processing:** Take the recommended configuration's capacity; express as a delta from the governing band.
- **Output:** `commercially_preferred_capacity` with the profile version that produced it.
- **Priority:** P1 (Phase 1)
- **Acceptance Criteria:** AC1: Always accompanied by the governing band, never shown alone. AC2: Delta from the governing band explained by named objective terms.
- **Dependencies:** FR-OBJ-001, FR-CAP-001.

9.3 Phase 2 — Precedent

FR-PRE-001 — Precedent Corpus with Tiering and Licensing

- **Requirement:** Structured corpus of approved projects, tiered by completeness, licence-tagged per record.
- **Rationale:** Copyright and licence exposure in a precedent corpus is a real diligence risk. Tiering is required because attribute data and geometry data have completely different availability.
- **Input:** Path A public data; Path B partner portfolios.
- **Processing:** Entity resolution to plot; normalization; derived metrics; tier assignment; licence tagging; regulatory-regime tagging.
- **Output:** `PrecedentProject[]`.
- **Priority:** P0 (Phase 2)
- **Acceptance Criteria:**
 - AC1: Every record carries source, licence, `redistributable` flag, acquisition date.
 - AC2: **Licence review is a gate on ingestion**, owned by a named person, not a cleanup task.
 - AC3: Records lacking a regulatory-regime tag cannot enter the similarity index.
 - AC4: Tier is explicit; Tier A (with geometry) is not required for the phase to succeed.
- **Dependencies:** §30 data acquisition.

FR-PRE-002 — Pattern Extraction with Minimum-n

- **Requirement:** Derive conditional distributions, never single values.
- **Rationale:** A precedent-derived point estimate is indistinguishable from a fact in the reader's mind. A distribution with n is not.
- **Input:** Corpus; conditioning context.
- **Processing:** Compute distributions with sample counts and intervals.
- **Output:** `PrecedentPattern` with n, median, quartiles, deciles.
- **Priority:** P0 (Phase 2)
- **Acceptance Criteria:**
 - AC1: No pattern emitted with $n < 8$; below that, return `INSUFFICIENT_PRECEDENT`.
 - AC2: Every displayed pattern shows n, median and interquartile range.
 - AC3: `PrecedentPattern` cannot be a constraint source — type-level, CI-asserted (Principle 1).
- **Dependencies:** FR-PRE-001.

FR-SIM-001 — Faceted Similarity with Regime Gate

- **Requirement:** Retrieve relevant precedents with per-facet scores and per-domain transferability.
- **Rationale:** A single "92% similar" figure is false precision that encourages copying the closest project — the exact behaviour the product must prevent. Two plots can be near-identical dimensionally and non-transferable regulatorily.
- **Input:** Subject plot, context, query domain.
- **Processing:** **Stage 1** hard regime gate — exclude precedents under a materially different regulatory regime for the queried parameters, and report the exclusion count. **Stage 2** faceted scoring across `spatial`, `dimensional`, `regulatory`, `programmatic`, `contextual`, with `temporal` as a modifier and facet weights configured **per query domain**.
- **Output:** Ranked precedents with facet vector, per-domain transferability, and explicit divergences.
- **Priority:** P0 (Phase 2)
- **Acceptance Criteria:**
 - AC1: **No single composite percentage is ever displayed as the headline.** Facet breakdown only.
 - AC2: Each precedent lists its top three divergences.
 - AC3: Regime-gate exclusions shown as a count so the user knows filtering occurred.
 - AC4: Architect evaluation: $\geq 70\%$ of top-5 judged genuinely relevant across 30 subject plots.
- **Dependencies:** FR-PRE-001.

FR-CAP-004 — Historically Observed Capacity

- **Requirement:** The fifth capacity concept — what comparable plots actually achieved — introduced only under the entry conditions of §5.4.
- **Rationale:** Regulation sets the ceiling; parking, structure, market and negotiation set the outcome. The gap is typically where the pro forma lives. But claiming to know it without data is worse than not claiming it.
- **Input:** `PrecedentPattern` set; subject context.
- **Processing:** Locate the conditioning cell; if $n \geq 8$ and backtest thresholds met, emit a distribution; otherwise `INSUFFICIENT_PRECEDENT`.
- **Output:** `historically_observed_capacity` as a distribution with `n`, class `OBSERVED`.
- **Priority:** P0 (Phase 2)
- **Acceptance Criteria:**
 - AC1: Never emitted below $n = 8$.
 - AC2: Both backtest coverage ($\geq 80\%$) **and** sharpness ($\text{median}(P90-P10)/P50 \leq 20\%$) required before the feature is enabled for a jurisdiction.
 - AC3: Always displayed alongside bands A/B/C, never replacing them.
 - AC4: Configurations exceeding the 95th percentile of precedent are flagged `OPTIMISTIC_VS_PRECEDENT` — **advisory only, never removed from the result set**, because a legitimately better design should be allowed to beat precedent.
- **Dependencies:** FR-PRE-002, FR-SIM-001.

9.4 Phase 3 — Extraction

FR-EXT-001 — Document Extraction to Candidate Records

- **Requirement:** LLM extraction producing candidates that are never directly authoritative.

- **Rationale:** Principle: an LLM may read, structure and narrate; it may never author a rule that reaches production.
- **Input:** Regulation, standards, RFP documents.
- **Processing:** Route structured content to a deterministic table parser; prose to the LLM under a strict schema; then deterministic normalization (units, parameter binding, type coercion, arithmetic cross-checks); emit `status = DRAFT`.
- **Output:** Candidate records + an `UnparsedStatements` queue.
- **Priority:** P0 (Phase 3)
- **Acceptance Criteria:**
 - AC1: Candidates are not evaluable. The evaluator refuses `status != APPROVED`.
 - AC2: 100% of statements are either extracted or queued. Silent drops are a P0 defect.
 - AC3: Every field carries page and bounding-box reference rendered as a source highlight.
 - AC4: Confidence labelled `CALIBRATED_CONFIDENCE` or `HEURISTIC_CONFIDENCE` (§6.5).
- **Dependencies:** FR-RUL-001.

FR-EXT-002 — Human Review and Approval Workflow

- **Requirement:** Route every candidate through qualified human review before publication.
- **Rationale:** Legal consequence requires human accountability with a name attached.
- **Input:** Candidate records.
- **Processing:** Review queue sorted by impact then confidence; side-by-side source; correction capture; approval; second-reviewer requirement for life-safety and planning parameters.
- **Output:** Approved versioned rules; correction telemetry.
- **Priority:** P0 (Phase 3)
- **Acceptance Criteria:**
 - AC1: Rules reaching production without a named approver: **0**.
 - AC2: Every correction persisted with original value, corrected value, user, timestamp — the primary extraction-quality signal.
 - AC3: Review throughput measured and reported; it is the real capacity limit on the knowledge base.
- **Dependencies:** FR-EXT-001, FR-GOV-001.

9.5 Phases 4-5 — Generation and Optimization

FR-GEN-001 — Massing, Core and Floorplate Generation

- **Requirement:** Generate massing, core configuration and floorplate within enforced constraints.
- **Rationale:** This is the output customers imagine when they hear "AI"; it is also the least trustworthy layer and therefore the last built.
- **Input:** `BuildableEnvelope`, typology library, core rules, occupant load.
- **Processing:** Parametric instantiation; core component sizing from cited rules; corridor topology; travel distance and exit separation evaluated geometrically as `EVALUATIVE_ONLY` rejections.
- **Output:** `Massing`, `CoreConfiguration`, `Floorplate`.
- **Priority:** P0 (Phase 4)
- **Acceptance Criteria:** AC1: Every core component dimension traces to a rule or a declared assumption. AC2: 100% pass invariants and independent validation. AC3: Occupant-load circularity resolved per §11.7 with a bounded fixpoint.
- **Dependencies:** Phase 0-3 complete.

FR-GEN-002 — Unit Planning and Typical Floor

- **Requirement:** Pack unit zones to a target mix within dimensional constraints.
- **Rationale:** The hardest component in the product and the one with no ground truth.
- **Input:** Floorplate, core, unit library, mix targets, dimensional standards.
- **Processing:** Constrained assignment over a discretized plate; geometry adjudicated by the deterministic kernel, never by the solver's own arithmetic.
- **Output:** TypicalFloor with labelled zones and computed efficiency.
- **Priority:** P0 (Phase 4)
- **Acceptance Criteria:**
 - AC1: 100% geometrically valid — no overlaps, façade access where required, every zone adjoins the corridor.
 - AC2: Efficiency computed from actual geometry, never assumed.
 - AC3: Where the mix is unachievable, the achievable mix is returned with quantified deviation — never a silent substitution.
 - AC4: Architect panel $\geq 70\%$ (§7.6). Below 60%, the phase does not ship.
- **Dependencies:** FR-GEN-001.

FR-OPT-001 — Multi-Objective Search

- **Requirement:** Replace enumeration with search **only where enumeration is demonstrated insufficient**.
- **Rationale:** Introducing NSGA-II and CP-SAT before enumeration has failed adds the largest performance and correctness risk in the system for no measured benefit.
- **Input:** Evidence that the configuration space cannot be enumerated within budget at required granularity.
- **Processing:** Evolutionary outer loop; solver inner loop; hard constraints as feasibility filters, **never as penalty terms**; surrogates and caching per §8.4.
- **Output:** Pareto front; ranking decoupled from search.
- **Priority:** P0 (Phase 5)
- **Acceptance Criteria:** AC1: Zero hard-constraint violations, independently verified. AC2: Options distinct in design space (§18.3). AC3: ≤ 10 min p95. AC4: Immutable-run reproducibility maintained.
- **Dependencies:** Phase 4 stable.

9.6 Governance (all phases)

FR-GOV-001 — Shared Asset Versioning and Approval

- **Requirement:** Every shared production asset is versioned, approval-controlled, changellogged, and triggers impacted-run notification.
- **Rationale:** Changing a default bay-area factor from 32 to 30 silently changes every customer's capacity. That is a production change with commercial consequences and currently has no approval path.
- **Input:** Proposed change to a shared asset.
- **Processing:** Proposal → impact analysis (which runs, which customers, what magnitude) → approval by the asset owner → publication with a new version → notification.
- **Output:** Versioned asset; changelog; impacted-run list.
- **Priority:** P0
- **Acceptance Criteria:**

- AC1: Covers rules, **default assumptions**, area definitions, typologies, objective normalizations, and deferred/non-mechanizable classifications (Principle 10).
- AC2: No shared asset changes without a named approver.
- AC3: Impacted runs identified within 24 h and their owners notified.
- AC4: Past runs are never mutated by an asset change.
- **Dependencies:** §13.4.

FR-GOV-002 — Objective Profile Governance

- **Requirement:** Objective weights are an organizational governance artifact, not a per-user preference.
- **Rationale:** If every user sets their own weights, two people in the same company produce contradictory recommendations for the same plot and the product's authority collapses at the investment committee.
- **Input:** Organization policy; project profile; exploratory adjustments.
- **Processing:** Three tiers — org policy (set by the development director, versioned, audited), project profile (within org bounds, justification required outside them), exploratory (watermarked, non-exportable). Role affects **presentation order only**, never ranking.
- **Output:** Versioned `ObjectiveProfile`.
- **Priority:** P0 (Phase 1)
- **Acceptance Criteria:** AC1: Every report states the profile ID and version. AC2: Divergent rankings by role are a defect, not a feature. AC3: Normalization functions are themselves versioned assets under FR-GOV-001.
- **Dependencies:** FR-OBJ-001.

FR-GOV-003 — Variance Handling

- **Requirement:** A documented, evidenced, expiring override of a rule for a specific plot.
- **Rationale:** Without an explicit variance mechanism, a user holding a legitimate approved exemption will edit the rule base to work around a correct refusal, silently corrupting a shared asset.
- **Input:** User assertion plus an evidence document.
- **Processing:** Create a `Variance` bound to rule + plot + evidence + asserter + expiry; second-user review where the tenant is multi-seat; apply to the constraint set with a prominent report annotation.
- **Output:** `Variance` record; modified constraint set.
- **Priority:** P1 (Phase 1)
- **Acceptance Criteria:**
 - AC1: **No variance without an uploaded evidence document.** No "upload later."
 - AC2: Life-safety rules cannot be varied — the pathway is disabled for that rule class.
 - AC3: Every report produced under a variance carries a face-page notice.
 - AC4: Variances expire and require re-confirmation.
 - AC5: Second-reviewer requirement is tenant-configurable and defaults on for multi-seat organizations — single-seat users are not blocked.
- **Dependencies:** FR-RUL-001.

10. Data Model

10.1 Design principles

Separate epistemic classes at the type level. `Rule`, `Standard`, `Target`, and `PrecedentObservation` are distinct types with distinct capabilities. `PrecedentObservation` is structurally incapable of being a `Constraint` source. Principle 1 is a property of the type system, not a policy someone can forget.

Everything authoritative is bitemporal. Validity time (when the rule was in force) and transaction time (when we recorded it). Without both, you cannot reproduce a past run or answer "was this correct at the time?"

Parameters are first-class. `setback.front` is an entity, not a string key. Rules, standards and targets all bind to it. This is what makes cross-source conflict detection tractable rather than string-matching.

Runs are immutable. Results are append-only. An asset change creates a new run; it never mutates an old one.

10.2 Core entities

```

METRIC_DEFINITION                                -- FR-DEF-001; root of everything
  metric_id, name, formula_statement
  inclusions[], exclusions[]
  version, approved_by, approved_at

PARAMETER                                        -- the shared vocabulary
  parameter_id                                  -- 'setback.front', 'parking.ratio.2bed'
  name, domain, unit_type, value_type
  is_life_safety: bool
  rule_class_default: GENERATIVE | FILTERING | EVALUATIVE_ONLY

PLOT
  plot_id, tenant_id
  plot_number, community, land_use
  stated_area, computed_area, area_mismatch_flag
  geometry: Polygon, geometry_source, geometry_class (USER_SET in Phase 0)
  shape_class: RECTILINEAR | SIMPLE_CONVEX | COMPLEX    -- COMPLEX rejected in Phase 0
  derived: principal_axis, mbr, convexity_ratio, frontage_count

PLOT_EDGE
  edge_id, plot_id, seq, geometry: LineString
  classification: ROAD | ADJACENT_PLOT | OPEN_SPACE | OTHER
  road_hierarchy, length_m, bearing_deg
  applied_setback_value, applied_setback_rule_id

RULE                                             -- §11.2
DEVELOPER_STANDARD                             -- Phase 1
PROJECT_BRIEF                                  -- Phase 1
OBJECTIVE_PROFILE                              -- Phase 1, versioned, governed

CONSTRAINT_SET                                 -- materialized, immutable per run
  generative[], filtering[], evaluative[], deferred[]
  preferences[]
  -- TYPE INVARIANT: no PrecedentObservation in any derivation path

BUILDABLE_ENVELOPE
  setback_permitted_footprint_m2, coverage_cap_m2, podium_footprint_m2
  tower_plate_cap_m2, height_ceiling_m
  max_levels_by_height, binding_constraint_per_dimension{}

CAPACITY_RESULT
  regulation_limited_gfa                       -- band A
  geometry_limited_gfa                         -- band B
  parking_limited_gfa                          -- band C
  governing_band, governing_constraint_rule_id
  commercially_preferred_gfa                   -- Phase 1, nullable
  historically_observed_gfa                    -- Phase 2, nullable, distribution
  user_realism_discount                        -- USER_SET, default 1.00
  integer_granularity_loss_m2

CONFIGURATION                                  -- Phase 1+
  config_id, run_id, typology_id, parameter_vector
  massing{}, core{}, unit_schedule[], parking{}
  metrics{}, scores{}
  invariant_report_id, validation_report_id, provenance_root_id

PRECEDENT_PROJECT                             -- Phase 2; §17.1
PRECEDENT_OBSERVATION                         -- NEVER a constraint source
VARIANCE                                       -- evidence-bound, expiring, non-life-safety
RUN                                             -- immutable; §13.4
PROVENANCE_NODE / EDGE                       -- §13.2
INVARIANT_REPORT                              -- §12

```

10.3 What changed from V1

Change	Reason
METRIC_DEFINITION added as a first-class entity	V1 used GFA/BUA/NSA/efficiency without definitions; the root cause of its broken example
shape_class on plot	Enforces the Phase 0 geometry restriction at the data layer, not in a comment
realistic_capacity field removed ; historically_observed nullable	Cannot be computed without data that does not yet exist
user_realism_discount added, USER_SET	Honest substitute during Phases 0-1

Change	Reason
<code>binding_constraint_per_dimension</code> promoted to a structured field	It is the most decision-relevant output and was buried in prose
<code>integer_granularity_loss</code> added	The floor that does not fit is real capacity loss and was previously absorbed silently
Rule <code>class</code> (generative/filtering/evaluative) added	V1 required all rules to be constructive, which is impossible
<code>INVARIANT_REPORT</code> added	New layer

11. Rule Model

11.1 Typed records, not a DSL

Decision: Phase 0 stores rules as typed data records evaluated by a registry of named functions. No DSL, no compiler.

Why this reverses V1. A DSL designed before ~100 rules exist encodes the wrong abstractions. Typed records deliver bitemporality, provenance, versioning and testability immediately, and leave genuinely awkward rules expressible as a registered Python evaluator rather than as a fight with a grammar. Extract a DSL around rule 100, when the shape is known. The DSL will be better and the migration is mechanical because the records are already typed.

Guard: the evaluator registry is capped at 12 functions in Phase 0. Exceeding it signals the abstraction is wrong and triggers a design review rather than a thirteenth function.

11.2 Rule record

```
{
  "rule_id": "R-SETBACK-ROAD-RES",
  "parameter_id": "setback.road",
  "rule_class": "GENERATIVE",
  "evaluator": "table_lookup_min",
  "evaluator_args": {"key": ["road_hierarchy"], "table": "T-SETBACK-01"},
  "unit": "m",
  "applicability": {
    "all": [
      {"land_use": {"in": ["RESIDENTIAL_MULTI"]}},
      {"edge.classification": {"eq": "ROAD"}}
    ]
  },
  "citation": {
    "instrument_id": "...", "instrument_version": "...",
    "clause_reference": "...", "document_uri": "...",
    "source_page": 0, "source_bbox": [0,0,0,0],
    "source_text_verbatim": "..."
  },
  "jurisdiction": "DM MAINLAND",
  "is_life_safety": false,
  "mechanization": "MECHANIZED",
  "valid_from": "YYYY-MM-DD", "valid_to": null,
  "recorded_at": "...", "version": 3, "supersedes": "...",
  "status": "APPROVED",
  "authored_by": "...", "approved_by": "...", "approved_at": "...",
  "tests": [
    {"kind": "POSITIVE", "context": {...}, "expected": 6.0},
    {"kind": "BOUNDARY", "context": {...}, "expected": 4.5},
    {"kind": "NEGATIVE", "context": {"land_use": "INDUSTRIAL"}, "expected": "NOT_APPLICABLE"}
  ]
}
```

All nine Principle-9 fields are non-null. `status != APPROVED` is not loadable by the evaluator.

11.3 The three rule classes

Class	Definition	Operational role	Examples
GENERATIVE	Can directly construct geometry or a numeric bound	Builds the envelope; runs first	Setbacks (polygon offset), height ceiling, FAR ceiling, coverage cap, tower plate cap
FILTERING	Prunes candidates during construction	Applied at instantiation; a candidate violating it is never created	Minimum unit area, minimum corridor width, minimum stair width, parking ratio

Class	Definition	Operational role	Examples
EVALUATIVE_ONLY	Assessable only after a candidate exists; has no constructive inverse	Applied post-construction; can only reject, never modify	Maximum travel distance, exit separation, dead-end corridor length, aggregate occupant load
DEFERRED	Applicable but not mechanizable at this fidelity	Declared in every output	Performance-based smoke control, façade fire performance, discretionary judgement

Why this matters for the architecture: V1 required every rule to declare a "constructive form," which is impossible for topological rules — there is no way to generate a floorplate *from* a travel-distance limit. **EVALUATIVE_ONLY** rules prune only after the fact, which changes search efficiency materially. Phase 5 must budget for rejection rates in the outer loop; that cost is now visible.

11.4 Evaluator registry (Phase 0, ≤12)

scalar_min · scalar_max · table_lookup_min · table_lookup_max · ratio_per_unit_type · percentage_of_base · polygon_offset_inward · count_minimum · range_bound · enum_permitted_set · conditional_scalar · custom_python (registered, tested, reviewed like any other rule).

11.5 Overlap resolution — scalar parameters

For a parameter with multiple applicable rules:

1. A plot-specific instrument specifying the parameter governs — it is the instrument that applies the general framework to this plot. **Exception: it cannot relax a life-safety parameter.**
2. Otherwise, across authorities, **most restrictive governs**. An overlay adds restriction; it does not relax a baseline.
3. Every non-governing candidate is recorded as a `supersededBy` edge and is visible to the user.

"Most restrictive" is well-defined for scalars: for **MIN** operators the maximum value; for **MAX** operators the minimum value.

11.6 Non-scalar conflict handling

"Most restrictive wins" is **undefined** for enumerated values, topological rules and incommensurable units. V1 asserted the rule without defining the order. The resolution:

Parameter type	Partial order	Resolution
Scalar with MIN/MAX	Total order	Automatic
Bounded range	Intersection	Automatic; empty intersection → IRRECONCILABLE
Count	Total order (max)	Automatic
Enumerated set	Set intersection	Automatic if intersection non-empty; empty → IRRECONCILABLE
Enumerated single value	No order exists	REQUIRES_ADJUDICATION
Topological	No order exists	REQUIRES_ADJUDICATION
Cross-unit (ratio vs. absolute)	Comparable only within a fixed context	Evaluate both in context; if still incomparable → REQUIRES_ADJUDICATION

REQUIRES_ADJUDICATION behaviour: the conflict is surfaced, computation for the affected parameter halts, and a human-authored `AdjudicationRecord` is required. The record is a shared, versioned,

approval-controlled asset under FR-GOV-001 — authored once per conflicting rule pair, reused across all projects, and cited in every report that relies on it.

IRRECONCILABLE behaviour: no configuration exists that satisfies both. The system reports infeasibility naming both rules — it does not pick one.

The system never silently resolves a conflict it has no defined order for. This was an unresolved gap in V1 and would have produced arbitrary behaviour at runtime.

11.7 Circular dependencies — the occupant-load fixpoint

The circularity is real: applicable rules depend on occupant load → occupant load depends on the design → the design depends on the rules.

Resolution: a bounded, conservative, monotone fixpoint.

1. SEED conservatively.
Occupant load seeded at the maximum the buildable envelope could physically support under the least restrictive plausible density. Conservative = over-estimates load = selects the more restrictive rule set. Erring toward more restriction is the safe direction.
2. ITERATE.
Compute rule set from current load → compute configuration → recompute load from the configuration.
3. CONVERGENCE TEST.
Converged when the applicable rule set is identical across two consecutive iterations. Note: on the RULE SET, not on the load value — rules switch at thresholds, so load can oscillate within a band while the rule set is stable.
4. BOUND.
Maximum 5 iterations.
5. NON-CONVERGENCE.
If the rule set has not stabilized in 5 iterations, the system does NOT pick one. It emits NON_CONVERGENT_APPLICABILITY, names the oscillating rules and the threshold they straddle, and requires the user to declare the occupancy basis (recorded USER_SET). Computation resumes from that declaration.
6. RECORD.
The iteration history is part of the provenance graph. The report states how many iterations were required.

Why the seed is conservative rather than optimistic: an optimistic seed can converge to a rule set that is less restrictive than the true one — a silent under-constraint. A conservative seed converging downward is safe at every step. Where the two directions disagree, the system reports the discrepancy rather than choosing.

12. Invariant Model

12.1 Why this layer exists

The V1 worked example claimed a 21-storey tower with a 792 m² plate and 6,700 m² GFA. $21 \times 792 = 16,632$. It claimed 172 units in 5,268 m² of NSA against minimum unit areas of 65–155 m² — 30.6 m² per unit. Its own contradiction checker, two pages later, computed a maximum near 55–60 units on the same plot.

Nothing caught any of it, because constraint checking asks "*is this permitted?*" and invariant checking asks "*does this arithmetic close?*" They are different questions. Only the second catches a configuration that is impossible rather than impermissible.

These are conservation laws, not constraints. They are cheap, they are absolute, and they run on every artifact the system produces — including documentation examples, test fixtures, and demo data.

12.2 Invariant catalogue

ID	Invariant	Tolerance
INV-01	$\Sigma(\text{level GFA}) = \text{total GFA}$	0.1%
INV-02	$\Sigma(\text{unit NSA}) + \text{core} + \text{circulation} + \text{services} = \text{level gross area}$	0.5%
INV-03	$\text{total GFA} / \text{plot area} = \text{reported FAR}$	0.1%
INV-04	$\Sigma(\text{unit counts by type}) = \text{reported unit count}$	exact
INV-05	$\Sigma(\text{mix shares}) = 1.000$	0.001
INV-06	$\text{Reported unit count} \times \text{weighted mean unit NSA} = \text{total NSA}$	1%
INV-07	$\text{Reported efficiency} = \text{NSA} / \text{GFA}$, at the definition scope stated	0.1%
INV-08	$\text{Podium footprint} \leq \text{setback-permitted footprint}$	exact
INV-09	$\text{Podium footprint} \leq \text{coverage cap}$	exact
INV-10	$\text{Tower plate} \leq \text{podium footprint AND} \leq \text{tower plate cap}$	exact
INV-11	$\Sigma(\text{level heights}) = \text{reported total height} \leq \text{height ceiling}$	0.01 m
INV-12	$\text{Parking bays} \times \text{bay area factor} \leq \text{available parking area}$	exact
INV-13	$\text{Parking bays} \geq \text{computed demand}$	exact
INV-14	$\text{Level count} \times \text{level height budget}$ is integer-consistent	exact
INV-15	Every area term used appears in the metric definitions annex	exact
INV-16	Governing capacity = min(A, B, C) and is labelled as such	exact
INV-17	$\text{Achieved FAR} \leq \text{permitted FAR}$	exact
INV-18	GFA composition matches the declared parking-in-FAR treatment	exact

12.3 Independence and enforcement

- Separate module. **No imports from the generator or capacity engine.** Enforced by an import linter in CI (Principle 4).
- Owned by an engineer who does not own the capacity engine.
- Failure **blocks emission**. Never a warning, never a configurable severity.

- Runs against every worked example in every product document as a CI job. **This document's Appendix example is verified by that job.**

12.4 What invariants are not

They do not establish compliance, correctness of rules, or fitness for purpose. A configuration can pass every invariant and be entirely non-compliant — it is merely arithmetically coherent. This distinction appears in the report (§16.5).

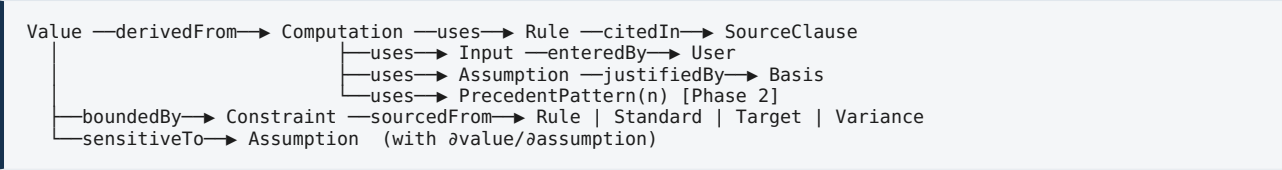
13. Provenance Model

13.1 Provenance classes

Class	Meaning	Rendering
DERIVED	Computed from a cited rule	Neutral, citation link
ASSUMED	System default filling a gap	Amber , dotted underline, inline-editable
USER_SET	Supplied or overridden by a named user	User badge
OBSERVED	From precedent distribution (Phase 2+)	Distribution sparkline with n
TRADEOFF	Optimizer choice among feasible alternatives (Phase 5)	Trade-off link showing what was given up
VARIANCE	Governed by a documented exemption	Red border, evidence link

The **amber ASSUMED** treatment is the most important UI decision in the product. The core failure mode is a user mistaking an assumption for a fact. Making assumptions visually loud, everywhere, permanently, is the mitigation. This must survive design review; softening it for aesthetic reasons defeats the product.

13.2 Graph shape



13.3 CI-asserted invariants on provenance

∀ emitted value	: provenance_class ≠ null
∀ DERIVED value	: ∃ cited rule in derivation path
∀ ASSUMED value	: ∈ assumption register ∧ sensitivity computed
∀ OBSERVED value	: ∃ pattern in path ∧ pattern.n ≥ 8
∀ constraint	: ∄ PrecedentObservation in derivation path
∀ numeral in prose	: ∈ values(provenance subgraph) [Phase 3+]
∀ life-safety rule	: ∄ variance

13.4 Immutable run reproducibility

V1 required "100% run reproducibility over 24 months." That is impossible — the pipeline includes LLM calls from Phase 3, models are deprecated, and even at temperature 0 batching and hardware introduce non-determinism. The requirement conflated two different things.

Corrected definition:

Term	Definition	Guarantee
Immutable run reproducibility	A run's stored outputs, inputs, versions, and provenance graph are retrievable unchanged, and the <i>deterministic portion</i> of the pipeline can be re-executed from persisted intermediate artifacts to produce byte-identical results	100%, required
Pipeline re-execution	Re-running the full pipeline including extraction from source documents	Not guaranteed. Produces a new run with a new ID.

What is persisted to make this hold: post-extraction structured inputs, rule set version and content hash, metric definitions version, all assumption values, all user inputs, engine version, RNG seed, and the full provenance graph.

The practical consequence for a customer: *"we can always show you exactly what produced this number, and we can always recompute it identically from what we stored — we cannot promise that re-reading your PDF today gives the same extraction as it did last year."* That is both honest and sufficient for every audit purpose the product serves.

14. Geometry Model

14.1 Phase 0 restriction

Rectilinear and simple convex polygons only. Enforced at the data layer via `shape_class`, not by convention. Complex and non-convex plots are rejected with a message naming the restriction. Lifted in Phase 4.

Why: partitioning a non-convex polygon into rectangles under simultaneous area, frontage, depth, aspect-ratio, façade-access and corridor-adjacency constraints is a research area, not a configuration option. It is tractable on rectilinear footprints via strip decomposition. Phase 0 does not need it and should not carry its risk.

14.2 Operations required in Phase 0

Per-edge inward offset · polygon intersection · area (with independent recomputation by a second method) · principal axis · minimum bounding rectangle · convexity ratio · point-in-polygon · polygon validity.

14.3 Numerical policy

- Exact predicates for orientation and intersection; never raw floating-point comparison
- Every area computed twice by independent methods; disagreement > 0.1% is a **P0 defect**, not a rounding note
- Snap tolerance declared explicitly, not implicit in the library
- Degenerate results (slivers, near-tangent offsets, self-intersections) raise rather than return

This is the class of bug that produces a plausible answer wrong by 4%, silently, for six months. The double-computation policy is the cheapest insurance available.

14.4 Coordinate handling

Local metric CRS (UTM 40N) for all computation. WGS84 stored for display only. No geometric operation in a geographic CRS.

14.5 Deferred to Phase 4

Non-convex and irregular plots · unit packing on arbitrary polygons · 3D massing operations beyond extrusion · façade surface analysis.

15. Capacity Model

15.1 Five distinct capacity concepts (Principle 6)

Never merged, never averaged, each with its own derivation and its own field.

#	Concept	Question	Available from	Provenance
A	Regulatory capacity	What does FAR and the area caps permit?	Phase 0	DERIVED
B	Geometric capacity	What does the envelope physically hold under height and footprint limits?	Phase 0	DERIVED
C	Parking capacity	What can the achievable parking supply support?	Phase 0	DERIVED
D	Commercially preferred capacity	What does the developer's own objective profile select?	Phase 1	DERIVED + profile version
E	Historically observed capacity	What have comparable plots actually achieved?	Phase 2, gated	OBSERVED, with n

15.2 Governing capacity

`governing = min(A, B, C)`, always named, always with its binding rule cited. Headroom to the next-binding constraint is reported, because that is what tells a developer where to push.

15.3 What is deliberately absent

There is **no "realistic," "expected," or "likely" band in Phase 0 or Phase 1**. Absence is enforced by schema — the field does not exist. In its place, `user_realism_discount`, defaulting to 1.00, class `USER_SET`, attributed to the user who entered it.

Why: the metrics that make a realism estimate credible — achieved-vs-permitted FAR, and efficiency — require *permitted* FAR per plot and NSA per building. Neither is available in public municipal permit data. Manufacturing the band from an unvalidated heuristic and labelling it "realistic" is the single most damaging thing this product could do, because it is the number customers will act on and it is the one they cannot check.

15.4 Integer granularity

Capacity is quantized by floors. When permitted GFA exceeds what an integer number of floors consumes, the remainder is **reported as `integer_granularity_loss`**, not absorbed. In the worked example this is 300 m² (1.88% of permitted GFA) — small, but the kind of number a developer wants to see rather than have smoothed away.

16. Validation Model

16.1 Three constraint categories

Category	Definition	Behaviour
HARD	Must not be violated	Feasibility filter; violating candidates are never emitted
SOFT	Preferred; deviation scored	Optimized; deviation reported
DEFERRED	Applicable but not checkable at this fidelity	Declared explicitly in every output

16.2 Contradiction battery (Phase 1)

Run before any computation:

target_gfa	vs	units × mix × min_unit_area / target_efficiency
target_unit_count	vs	band_A_gfa / weighted_mean_unit_area
target_floors	vs	height_ceiling / min_floor_to_floor
parking_implied_area	vs	available_levels × footprint × usable_fraction
target_sellable	vs	target_gfa × max_plausible_efficiency
target_gfa	vs	band_A_gfa

Each failure names the incompatible pair and quantifies the shortfall.

16.3 Independence

Validator shares no evaluation code with the generator; separate module, separate owner, import-linted.

16.4 The honest limit of independence

Implementation independence is not semantic independence. The validator evaluates **the same encoded constraint set the generator consumed**. It catches generator implementation bugs. It does not catch a mis-encoded rule, a missing rule, or a rule whose applicability predicate is wrong.

V1 stated a "0% hard-constraint violation" metric as though it meant compliance. It means self-consistency. This document says so wherever the metric appears.

16.5 The five-way claim statement

Every validation report contains these five statements, separately, verbatim (Principle 8):

SELF-CONSISTENCY	The configuration satisfies every constraint we encoded, and its arithmetic closes. Status: PASS / FAIL
RULE COVERAGE	We encoded N of M requirements identified as applicable. K are deferred and listed below. Status: N/M, coverage X%
GEOMETRIC VALIDITY	Geometry is topologically valid; areas independently recomputed and agree. Status: PASS / FAIL
REGULATORY VALIDITY	NOT ASSESSED. This system does not and cannot determine whether an authority would approve this scheme.
PROFESSIONAL AGREEMENT	Where measured against qualified architects on comparable plots, this engine fell within the inter-architect disagreement band in X of Y cases. Status: <measured value or NOT YET MEASURED>

The fourth line is permanent and never changes. The fifth is populated from the golden set (\$23) and is the honest answer to "is it accurate?"

17. Historical Precedent Model

17.1 Precedent record and tiering

```
PRECEDENT_PROJECT
precedent_id
acquisition_path: PUBLIC | PARTNER
data_tier: A | B | C
source, licence, redistributable: bool, acquired_at
plot_reference, community, approval_date, land_use
plot_area, permitted_far, built_up_area, gfa
floors_above, floors_below, unit_count, unit_mix, nsa
parking_bays
achieved_far (derived), efficiency (derived, requires nsa)
regulatory_regime_id -- required for the similarity gate
geometry -- Tier A only
feature_vector
```

Tier	Contents	Enables
A	Attributes + geometry + approval history	Massing precedent, layout patterns
B	Full attributes incl. NSA, mix, permitted FAR	Efficiency and achieved-vs-permitted calibration — the fields that matter
C	Partial attributes (typical of public data)	Density, floor count, parking ratio benchmarking only

The honest position: public municipal permit data yields Tier C, occasionally Tier B-minus. It does not carry NSA, unit mix, or *permitted* FAR. Tier B comes overwhelmingly from partner portfolios. \$30 plans both paths accordingly.

17.2 Precedent is evidence, never authority

Enforcement is structural, at three levels:

- 1. **Type level** — `PrecedentObservation` cannot be a `Constraint` source. Not a runtime check; a type error.
- 2. **CI assertion** — no precedent node appears in any constraint's derivation path.
- 3. **Output linting** — report text framing precedent as permissive ("this is allowed because a previous project...") is rejected by a linter before render.

17.3 What precedent is for, in priority order

- 1. **Calibration** — converting regulatory ceilings into observed outcomes
- 2. **Plausibility bounding** — catching the system's own optimism, flagged advisory-only
- 3. **Benchmarking** — where does this configuration sit against comparable approvals
- 4. **Generator seeding** (Phase 4) — biasing typology selection toward what is actually built
- 5. **Explanation** — "34 comparable approvals, median efficiency 76%" is the sentence that persuades a committee

Not for: overriding regulation, geometric copying, or justifying a non-compliant scheme.

17.4 Similarity measurement — the definition

Stage 1 — Regime gate (binary, hard). Exclude precedents governed by a materially different regulatory regime *for the parameters being queried*. A project approved under a superseded

parking standard is not comparable for parking questions, however similar its plot. The exclusion count is reported so the user knows filtering occurred.

Stage 2 — Faceted scoring. A vector, never a scalar.

Facet	Components
spatial	distance, same community, same sub-community
dimensional	plot area ratio, aspect ratio, shape complexity, frontage count
regulatory	land use, permitted FAR, height band, coverage cap
programmatic	building type, unit mix, density
contextual	orientation, adjacency, road hierarchy
temporal	approval recency (decay modifier)

Facet weights are per query domain, not global. For a parking question, regulatory and temporal dominate and contextual is nearly irrelevant. For a massing question, dimensional and contextual dominate.

Stage 3 — Transferability, which is not similarity. Similarity measures how alike two plots are. Transferability measures how much a *specific lesson* survives the difference. A precedent 88% similar overall but with two road frontages instead of one has high transferability for efficiency ratios and low transferability for massing and access strategy. Reported per domain, with the divergences that reduce it.

Presentation rule: never a headline percentage. Facet breakdown, top-three divergences, and a plain-language transferability statement. A "92% similar" badge is false precision and encourages exactly the copying behaviour Principle 1 forbids.

18. AI Architecture

18.1 The deterministic / AI boundary

Concern	Implementation	Phase
Geometry, areas, envelopes	Deterministic kernel	0
Rule applicability and evaluation	Deterministic	0
Capacity computation	Deterministic	0
Parking demand	Deterministic	0
Invariants	Deterministic	0
Validation	Deterministic, independent module	0
Configuration enumeration and scoring	Deterministic	1
Similarity scoring	Deterministic faceted (learned facet weights only much later)	2
Document extraction	LLM/VLM , schema-constrained, human-reviewed	3
Table extraction	Deterministic parser — not the LLM	3
Unit normalization, parameter binding	Deterministic pass over model output	3
Requirement classification	Deterministic mapping once the parameter is identified	3
Rule authoring	Human-authored, LLM-drafted, never auto-published	3
Explanation prose	LLM, graph-grounded, numerically post-verified	3
Unit packing	Constraint solver	4
Massing search	Enumeration → search only if justified	1 → 5

18.2 The invariant

The LLM may read, structure, and narrate. It may never compute a number that reaches a user, and it may never author a rule that reaches production.

18.3 Design-space distinctness (not objective-space)

V1 required generated options to be "distinct in objective space above a threshold." That is the wrong space: two schemes can score far apart and be formally identical, or score nearly identically and be completely different buildings.

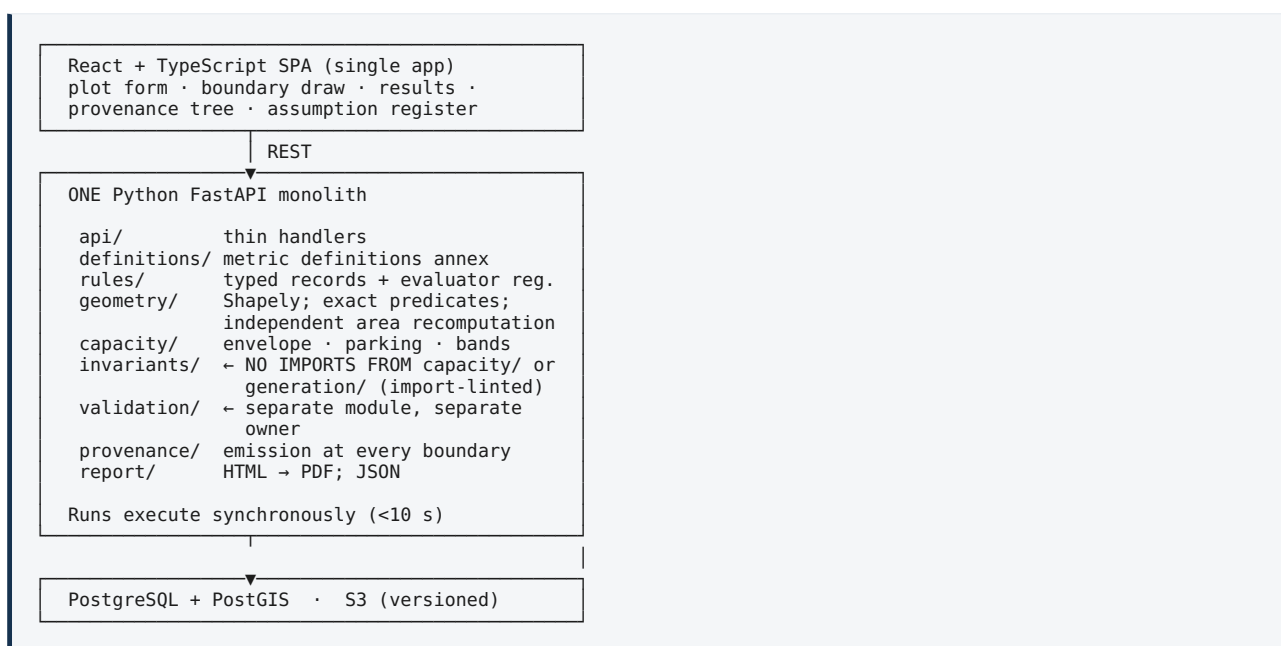
Corrected: distinctness is measured in **design space** — typology identifier, core typology, tower plate aspect ratio (binned), floor count, podium level count. Two configurations are distinct if they differ in typology or core type, or in ≥ 2 of the remaining dimensions. The threshold is a versioned asset under FR-GOV-001.

18.4 Why not end-to-end generative

Non-reproducible, unexplainable, unfixable (a wrong setback can only be retrained, not corrected), untestable per rule, and legally indefensible. It would also require a training corpus of approved Dubai residential drawings that does not exist and could not be lawfully assembled.

19. Technical Architecture

19.1 Phase 0 architecture



19.2 Explicitly removed from Phase 0

Removed	Why	Reintroduce when
Microservices	Eight services for a team of four pre-PMF is a platform, not a product	Team > 12, or independent scaling need
Temporal / durable workflow	Phase 0 runs are seconds, not minutes	Phase 5, when runs are long and resumable matters — with run boundaries already understood
Redis	Nothing to cache at this latency	When a measured cache-hit benefit exists
pgvector	No embeddings until Phase 2 similarity	Phase 2, if vector search beats faceted scoring on the retrieval eval
Graph DB	Provenance is a bounded per-run DAG queried by root; Postgres recursive CTEs handle it	If cross-run graph queries become a product feature
Rule DSL + compiler	Premature abstraction before rule 100	~Rule 100, when the shape is known
NSGA-II / pymoo	No search until enumeration is proven insufficient	Phase 5, with evidence
CP-SAT / OR-Tools	No packing until Phase 4	Phase 4
Rhino / Grasshopper	Windows-licensed, stateful, hard to containerize; wrong in a SaaS critical path	Never in the core path; optional interop only
DXF export	JSON + PDF suffice for Phase 0	Phase 1

19.3 Retained from V1 because retrofitting is a rewrite

Bitemporal rule storage · immutable runs · provenance emitted at every layer boundary · validator as a separate module with a different owner · exact geometric predicates with independent area recomputation · invariant layer (new, but same argument).

These are cheap now and unaffordable later. Everything else can wait.

19.4 Stack by phase

Phase	Additions
0	Python 3.12, FastAPI, Shapely/GEOS, pyproj, PostgreSQL 16 + PostGIS, S3, React/TS, WeasyPrint, pytest
1	Background jobs (simple queue), DXF via ezdxf, XLSX via openpyxl
2	pgvector <i>if</i> it beats faceted scoring on the retrieval eval; data pipeline tooling
3	LLM API (zero-retention), layout-aware document parser, deterministic table parser, review UI
4	OR-Tools CP-SAT, 3D rendering (three.js), geometry kernel extension
5	pymoo, Temporal, Redis, autoscaling worker pool, surrogate model training

20. UX Flow

20.1 Phase 0 flow

1. PLOT	form + boundary draw + edge classification	
2. PARAMETERS	FAR, height, coverage, plate cap + parking-in-FAR declaration (blocking, no default)	
3. RULES	applicable set, citations, inclusions/exclusions	[ack gate]
4. ASSUMPTIONS	register, sensitivity-ranked, editable	[ack gate]
5. CAPACITY	bands A/B/C, governing band, binding constraints	
6. PARKING	demand, area, levels, headroom	
7. CHECKS	invariants · validation · five-way claim statement	
8. EVIDENCE	provenance tree, click-through from any number	
9. EXPORT	PDF + JSON	[reviewer gate]

Steps 1–4 are sequential and gated. Steps 5–9 are freely navigable.

20.2 The three moments that carry the product

1. **The parking-in-FAR declaration (step 2).** A blocking question with no default. It signals immediately that the system will not guess on things that matter.
2. **The assumption register (step 4).** The moment the user understands this is not magic. Non-skippable, sensitivity-ranked, inline-editable.
3. **Click-through provenance (everywhere).** Every number is a link. This is the interaction that converts a skeptical architect.

20.3 Uncertainty as a first-class visual state

State	Treatment
DERIVED	Neutral, citation superscript
ASSUMED	Amber background, dotted underline, pencil affordance
USER_SET	User badge
OBSERVED	Distribution sparkline with n
VARIANCE	Red border, evidence link
Deferred / not assessed	Grey, hatched, "not assessed" label

Uncertainty is not a tooltip.

20.4 Role lenses (Phase 1)

Role affects **presentation order only**. A development director sees GFA, unit count and value proxy first; an architect sees efficiency, aspect ratios and core relationships first. **Same computation, same ranking**. Divergent rankings by role would be a defect.

21. Human Approval Gates

21.1 Run gates

Gate	Actor	Blocks	Phase
G1 Plot & parameter confirmation	User	Rule resolution	0
G2 Rule set acknowledgement	User	Capacity computation	0
G3 Assumption register acknowledgement	User	Any export	0
G4 Named reviewer on export	User	External sharing	0
G5 Brief contradiction resolution	User	Configuration enumeration	1
G6 Variance review	Second user (tenant-configurable; default on for multi-seat)	Variance application	1

21.2 Shared asset gates (Principle 10)

The gates V1 missed. These have higher blast radius than run gates, because they affect every customer at once.

Asset	Approver	Why it needs a gate
Rule publication	Qualified regulatory architect	Legal consequence
Life-safety rule	+ fire/life-safety specialist (second reviewer)	Highest-consequence class
Default assumption values	Regulatory architect + product	Changing a default bay-area factor silently changes every customer's capacity
Metric definitions annex	Architect + design partner sign-off	Root of every computed value
Typology library entries	Regulatory architect	Each is a design assertion shipped to every customer
Objective normalization functions	Product + development director	Determines rankings
Deferred / non-mechanizable classification	Regulatory architect	Determines what the report claims <i>not</i> to have checked — a liability statement
Adjudication records (§11.6)	Regulatory architect	Resolves conflicts that have no defined order
Design-space distinctness threshold	Computational design lead	Determines what users see as "different options"

Every one is versioned, changellogged, and triggers impacted-run identification within 24 h. **Past runs are never mutated.**

22. Evaluation Framework

22.1 Layers

Layer	What	Method	Phase	Gate
E0 — Invariants	Arithmetic coherence	Automated, every artifact incl. docs	0	100%
E1 — Rules	Rule correctness	≥ 3 tests each, boundary + negative	0	100% pass
E2 — Geometry	Envelope and area correctness	Golden set, independent recomputation	0	$\leq 0.1\%$ self, band-match vs. architects
E3 — Professional agreement	Does a qualified architect agree?	Two-architect golden set (§23)	0	8/10 within band
E4 — Extraction	Field accuracy	Held-out gold set	3	$\geq 95\%$ critical
E5 — Precedent backtest	Do estimates match outcomes?	Held-out approved projects	2	Coverage $\geq 80\%$ and sharpness $\leq 20\%$
E6 — Generation quality	Are the floors any good?	Blind architect panel with real-floor controls	4	$\geq 70\%$, $\alpha \geq 0.6$
E7 — Decision impact	Did it change the outcome?	Customer-reported carry-through	1+	$\geq 40\%$ baseline

22.2 The inter-architect variance measurement — do this first

Week 1 of Phase 0, before any engine code. Give three qualified architects the same three plots with the same data packs and ask for buildable envelope, permitted GFA, and parking demand.

Expected spread on setback interpretation, edge classification, and projection treatment: **4-8%**. If that is what it is, then a 2% accuracy target against a single architect's answer is measuring noise, and the product claim changes from "*accurate*" to "*consistent, explicit, and reproducible*" — which is a stronger and more defensible claim anyway, and it should be measured before it is marketed.

22.3 Metrics that are absolute

ID	Metric	Target
M-INV	Emitted artifacts passing invariants	100%
M-PRV	Emitted values with a provenance class and complete derivation	100%
M-APP	Rules reaching production without a named approver	0
M-PRE	Precedent nodes in constraint derivation paths	0
M-DEF	Reports listing their deferred checks	100%
M-RUN	Immutable-run reproducibility (§13.4 definition)	100%

These are integrity properties, not quality targets. Any failure is a release blocker.

22.4 Metrics that must be paired

Metric	Must be paired with	Why
Backtest coverage $\geq 80\%$	Sharpness $\leq 20\%$	Coverage alone is gamed by widening the band
Self-consistency PASS	Rule coverage %	Passing constraints you did not encode is meaningless
Extraction accuracy	Queue-vs-drop rate	99% accuracy on 60% of statements is not 99% accuracy

23. Golden Dataset

23.1 Composition (Phase 0)

Item	Quantity	Source	Purpose
Real plots, one community, one land use	10	Design-partner land packs	Golden set
Independent architect computations per plot	2	Two architects, blind to each other	Ground truth and variance measurement
Mechanized rules	~40	Authored by the project architect	Rule base
Rule test cases	≥120	Concurrent with authoring	E1
Metric definitions	1 annex	Architect + partner sign-off	Root dependency

Total external data dependency: ten land packs from one friendly developer. No municipal negotiation. No open-data licensing. No scraping.

23.2 Why two architects, not one

A single architect's answer is not ground truth — it is one sample from a distribution whose width nobody has measured. Two independent computations give:

1. **A band, not a point.** The engine target becomes "within the band," which is an honest and defensible claim.
2. **The variance figure itself**, which is a headline product asset. If architects disagree by 6%, "we are consistent and reproducible" is a stronger pitch than "we are accurate."
3. **Detection of engine bias.** Systematically landing at one architect's end of the band is a signal worth investigating.

23.3 The circularity, acknowledged

The golden set requires FAR, height and setbacks per plot — the same plot data the engine needs and which may not be on the affection plan. **This is why the golden set comes from design-partner land packs, not from public sources.** It also makes the golden-set exercise a direct test of assumption A1 (§27): of ten land packs, how many actually contained FAR, height and setbacks without further enquiry? That number is a Phase 0 deliverable in its own right.

23.4 Rule test requirements

Every rule: ≥1 positive, ≥1 boundary (at the threshold), ≥1 negative (applicability correctly excludes). Boundary cases matter most — threshold-dependent rules switching at the wrong point is the highest-frequency rule defect.

24. MVP Acceptance Criteria

"MVP" here means **Phase 0 complete and shipped to design partners**. Phase 1 acceptance is §4.4.

24.1 Functional

- [] Metric definitions annex signed, versioned, cited in every report
- [] Parking-in-FAR treatment declared per project; no default; blocking
- [] Plot entry with boundary draw and mandatory edge classification, ≤ 3 min
- [] ~40 rules as typed records, all nine Principle-9 fields non-null, all approved by a named human
- [] Every rule classified generative / filtering / evaluative-only / deferred
- [] Buildable envelope with binding constraint named per dimension
- [] Parking demand $\rightarrow$ area $\rightarrow$ levels $\rightarrow$ headroom, with cited ratios
- [] Capacity bands A, B, C with derivations; governing band named
- [] Integer granularity loss reported
- [] User realism discount, default 1.00, `USER_SET`
- [] Assumption register, sensitivity-ranked, inline-editable, non-skippable
- [] Invariant layer: 18 checks, independent module, blocks emission
- [] Independent validation with the five-way claim statement
- [] Provenance graph, click-through from any number
- [] Deferred-check list in every output
- [] PDF + JSON export; JSON round-trips
- [] Immutable-run reproducibility per §13.4

24.2 Quality

- [] M-INV, M-PRV, M-APP, M-DEF, M-RUN all at 100% / 0
- [] Independent area recomputation agrees $\leq 0.1\%$ on all golden plots
- [] Engine within the two-architect band on ≥ 8 of 10 golden plots
- [] Full computation < 10 s
- [] Every worked example in every product document passes the invariant layer in CI

24.3 Measurement deliverables (equal weight to the software)

- [] Inter-architect variance measured and published
- [] Mechanization rate on the first 40 rules measured
- [] Rule authoring throughput measured, by complexity class
- [] Land-pack completeness rate measured (of 10, how many carried FAR/height/setbacks)
- [] Provenance-tree engagement observed in ≥ 3 user sessions

24.4 Governance

- [] Every shared asset versioned with a named approver
- [] Import linter enforcing invariant and validator module independence, in CI
- [] Professional-use disclaimer reviewed by UAE counsel

- ☐ No marketing material claiming compliance verification, guarantee, or assurance
-

25. Risk Register

ID	Risk	Likelihood	Impact	Mitigation	Owner
R-01	Team builds generation first because it demos well	High	Critical	Phase gating with dependency argument (§1.3); Phase 0 must ship before Phase 1 starts	Product
R-02	Land packs lack FAR / height / setbacks	High	High	Measured in Phase 0 as a deliverable; blocking OPEN state rather than silent default; if <5/10, the product's first job is data assembly — a different product	Product
R-03	Inter-architect variance > 10%	Medium	High	Measured week 1; if high, reposition from "accurate" to "consistent and auditable" before marketing	Product
R-04	Mechanization rate < 50%	Medium	High	Measured on first 40 rules; re-scope to a smaller, better-defended parameter set	Reg. Architect
R-05	Rule authoring slower than 4/day on complex rules	High	High	Measured; the first 40 rules run 3–5× slow because schema is co-designed. Budget 14–18 architect-weeks for 300 rules, not 6–10	Reg. Architect
R-06	Public precedent data lacks NSA, mix, permitted FAR	High	High	Stated as expected (§17.1); Path B partner portfolios are the primary Tier B source; INSUFFICIENT_PRECEDENT degrades gracefully	Data Lead
R-07	Professional liability	Low	Critical	Disclaimers, PI insurance before first paying customer, human gates, five-way claim statement, marketing review on the same footing as product	CEO / Counsel
R-08	Precedent corpus copyright exposure	Medium	High	Licence gate on ingestion owned by a named person; no scraping of marketed plans	Data Lead
R-09	Nested optimizer misses runtime budget by 10×	High	Medium	Enumeration in Phase 1 defers the problem entirely; §8.4 costs it honestly before Phase 5 design is chosen	Comp. Design Lead
R-10	Geometric degeneracy produces a plausible wrong answer	Medium	High	Exact predicates; double area computation ; degenerate results raise rather than return; invariant layer	Eng Lead
R-11	Users read self-consistency as compliance	High	Critical	Five-way claim statement verbatim in every report; marketing prohibition; disclaimer	Product
R-12	Assumption treated as fact	High	High	Amber everywhere, non-skippable register, quantified sensitivity	Design
R-13	Regulatory architect hiring is the schedule's binding constraint	High	High	Recruit before engineering ramp; founding-team role, not a hire	CEO
R-14	Shared-asset change silently alters every customer's numbers	Medium	High	FR-GOV-001; impacted-run notification; past runs never mutated	Eng Lead
R-15	Non-scalar conflicts resolved arbitrarily at runtime	Medium	Medium	§11.6: REQUIRES_ADJUDICATION halts rather than guesses	Reg. Architect
R-16	Occupant-load fixpoint fails to converge	Low	Medium	§11.7: bounded at 5 iterations, conservative seed, user declaration on non-convergence	Comp. Design Lead
R-17	Road hierarchy layer unavailable for automated edge classification	Medium	Low	Phase 0 makes edge classification manual; automation is a Phase 3+ convenience, not a dependency	Data Lead

ID	Risk	Likelihood	Impact	Mitigation	Owner
R-18	Customers reject Phase 0 as "just a calculator"	Medium	High	Tested in Phase 0 with design partners; the CEO brief (§31) is the counter-argument; if true, accelerate Phase 1	CEO

26. Kill / Re-scope Criteria

Agreed in advance so they are evidence-triggered, not argued.

Finding	Threshold	Action
Inter-architect variance	> 10%	Accuracy is not the value proposition. Reposition around consistency, explicitness and auditability. Rewrite marketing before shipping.
Mechanization rate on first 40 rules	< 50%	The compliance surface is too thin to claim coverage. Re-scope to a narrower, better-defended parameter set and reduce the claimed rule count publicly.
Rule authoring throughput on complex rules	< 4/day	Knowledge-base cost is ~3x budget. Either raise on that basis or narrow the jurisdiction slice further. Do not proceed on the original estimate.
Land packs containing FAR + height + setbacks	< 5 of 10	The input model is wrong. The product's first job is data assembly, not computation — that is a different product and needs a different plan.
Users opening the provenance tree	< 1 of 3 sessions	The trust thesis is wrong. Everything downstream of §13 needs re-examination before Phase 1.
Engine within two-architect band	< 6 of 10	Do not ship. Diagnose whether it is rules, geometry, or definitions. Fixing the wrong one wastes a quarter.
Phase 2 backtest coverage or sharpness	Either missed	Historically-observed capacity does not ship. Keep the user-entered discount.
Phase 4 architect panel	< 60%	Phase 4 does not ship. Phases 0–3 remain the product. Pre-authorized.
Phase 1 enumeration insufficient at production granularity	Not demonstrated	Phase 5 search does not begin. Enumeration stays.

27. Open Questions

ID	Question	Owner	Needed by	Consequence if unresolved
Q1	Does parking count toward FAR in the target jurisdiction slice?	Reg. Architect	Phase 0 week 2	Changes capacity 15–35%. Currently OPEN REGULATORY QUESTION and blocking per FR-DEF-002.
Q2	What are typical FAR values for target plots?	Product	Phase 0 week 1	If FAR is 3–5, a 2,000 m ² plot is a small point block, not a tower — and much of the generation machinery targets the wrong problem size
Q3	Inter-architect variance on envelope and capacity?	Reg. Architect	Phase 0 week 2	Sets the accuracy target and the marketing claim
Q4	What fraction of the governing residential-tower rule set is mechanizable?	Reg. Architect	Phase 0 week 8	Sizes the whole business
Q5	Do land packs actually contain FAR, height and setbacks?	Product	Phase 0 week 3	Tests assumption A1
Q6	Does a road-hierarchy GIS layer exist and join cleanly to cadastral edges?	Data Lead	Phase 3	Determines whether edge classification can ever be automated
Q7	Will developers pay for Phase 0 capacity alone?	CEO	Phase 0 week 10	Validates or invalidates the phase-gating strategy
Q8	Do public municipal datasets carry NSA, unit mix, or permitted FAR?	Data Lead	Phase 2 start	Determines whether Path A can support calibration at all — expected answer is no
Q9	What is the real distribution of achieved vs. permitted FAR?	Data Lead	Phase 2	The core calibration parameter
Q10	Is exhaustive enumeration sufficient at production granularity?	Comp. Design Lead	Phase 1 end	Determines whether Phase 5 is needed at all
Q11	What PI insurance is obtainable, at what premium, for this description of product?	CEO	Phase 0 week 8	Could change positioning or pricing
Q12	How frequently do the governing instruments actually change?	Reg. Architect	Phase 1	Sizes maintenance and the durability of the moat
Q13	Do customers accept a set of options, or demand a single answer?	Product	Phase 1	Shapes the recommendation UX

28. Data Acquisition Plan

28.1 By phase

Phase	Needed	Source	Difficulty	Blocking?
0	10 land packs, one community	Design partner	Low	Yes — but one relationship
0	Regulatory source documents	Published	Low	Yes
0	Basemap tiles	Commercial	Low	No
0	2× architect computations ×10	Contracted architects	Low	Yes
1	Developer standards documents	Customer	Low	No
1	Unit-type dimensional library	Architect-authored	Medium	Yes (Phase 1)
1	Typology library (4–6)	Architect-authored	Medium	Yes (Phase 1)
2	Public permit / building datasets	Municipal open data, commercial licence	Medium	No
2	Partner project portfolios	2–3 developer partners	High (relationship)	Yes for Tier B
2	Cadastral plot geometry	Licensing	Medium	No — manual fallback
3	Extraction gold sets	Internal labelling	Medium	Yes (Phase 3)
4	Approved floor plates (Tier A)	Partner or municipal	Very high	No — Phase 4 does not depend on it

28.2 Minimum viable precedent dataset (Phase 2)

Path A target — breadth. For ≥300 Dubai residential towers approved since 2010, ten fields: plot reference, community, approval date, land use, plot area, built-up area, floors above, floors below, unit count, parking bays. Sufficient for density, floor-count and parking-ratio benchmarking. **Not sufficient for efficiency or achieved-vs-permitted FAR.**

Path B target — depth. From 2–3 developer partners, 40+ projects with additionally: permitted FAR, NSA, unit mix, achieved efficiency. **This is the only realistic source of Tier B**, and Tier B is what makes the historically-observed capacity concept credible.

If Path B fails: the historically-observed band does not ship. Users keep the `USER_SET` realism discount. Benchmarking still works on Path A fields. The product degrades; it does not break. This is designed-in, not improvised.

28.3 What to ask a municipality for, in order

Tier 1 (likely obtainable): bulk/API access to published permit, building summary, floor-level, building usage and project datasets with a commercial licence permitting derivative analytical use; schema documentation; historical depth; licence clarification for a commercial derivative product; plot-level cadastral geometry under licence.

Tier 2 (requires relationship, transformative): approved GFA and permitted FAR per project; unit schedules and mix; structured setback and height conditions per plot; aggregated approval-cycle statistics; anonymized variance/exemption grant statistics by parameter.

Tier 3 (long horizon, do not plan around): approved drawing files; review comment text; submission→revision→approval sequences.

Do not make any roadmap contingent on Tier 2 or Tier 3.

28.4 Licensing discipline

Licence review is a **gate on ingestion**, owned by a named person, before any record enters the corpus — not a cleanup task. Marketed floor plans, broker listings and consultant drawings are protected works. Scraping them creates exposure that surfaces at diligence.

29. Corrections Applied from Red Team

#	Issue	Resolution	Section
1	"100% run reproducibility" impossible	Replaced with immutable run reproducibility — outputs and provenance always retrievable and recomputable from persisted artifacts; pipeline re-execution explicitly not guaranteed	§13.4
2	No invariant layer	Added: 18 conservation checks, independent module, blocks emission, runs on documentation examples	§12
3	Area terminology undefined	MetricDefinition annex as FR-DEF-001, signed week 1, root dependency, cited in every report	§9.1, §10.2
4	Parking-in-FAR unstated	Explicit OPEN REGULATORY QUESTION ; blocking user declaration; both scenarios computable; never ASSUMED	FR-DEF-002
5	"Realistic band" without data	Removed from Phases 0–1 by schema. Replaced with USER_SET discount. Reintroduced in Phase 2 only under $n \geq 8$ + coverage + sharpness gates	§15.3, §5.4
6	DSL premature	Typed rule records + evaluator registry capped at 12; DSL extraction at ~rule 100	§11.1
7	Over-engineered infrastructure	Microservices, Temporal, Redis, pgvector, graph DB removed from Phase 0 with named reintroduction conditions	§19.2
8	Geometry generality assumed	Phase 0 restricted to rectilinear/simple convex, enforced via shape_class at the data layer	§14.1
9	NSGA-II / CP-SAT premature	Removed from Phases 0–1; enumeration instead; introduced in Phase 5 only on evidence	§4.3, §8.3
10	"All rules constructive" impossible	Three rule classes: GENERATIVE , FILTERING , EVALUATIVE_ONLY , plus DEFERRED	§11.3
11	Occupant-load circularity unresolved	Bounded monotone fixpoint: conservative seed, convergence on rule-set identity, 5-iteration bound, user declaration on non-convergence	§11.7
12	Non-scalar conflicts undefined	Partial order defined per parameter type; REQUIRES_ADJUDICATION halts and demands a versioned human adjudication record; IRRECONCILABLE reports infeasibility	§11.6
13	Similarity undefined	Three stages: regime gate, per-domain faceted vector, transferability. No composite headline percentage.	§17.4
14	Confidence uncalibrated	Two labels: CALIBRATED_CONFIDENCE (may gate) and HEURISTIC_CONFIDENCE (display only, never gates)	§6.5
15	No shared-asset gates	FR-GOV-001 covering rules, defaults, definitions, typologies, normalizations, deferred classifications, adjudications, distinctness thresholds	§21.2
16	Worked examples arithmetically invalid	All V1 examples withdrawn. New example derived from first principles and CI-verified against the invariant layer	Appendix A
17	Backtest coverage gameable	Paired with a sharpness requirement	§5.4, §22.4
18	Distinctness in objective space	Corrected to design space	§18.3
19	Validator independence oversold	§16.4 states the honest limit; five-way claim statement replaces "0% violations = compliant"	§16.4, §16.5

Appendix A — Worked Example (Invariant-Verified)

Every number below was computed by the invariant checker and passes all applicable checks in §12.2. The verification script is part of the CI suite. Figures are illustrative of the *method*; parameter values are marked with their provenance class.

A.1 Inputs

Item	Value	Class
Plot dimensions	80.0 × 40.0 m	USER_SET
Plot area	3,200.0 m ²	DERIVED (computed, matches stated)
Shape class	RECTILINEAR	DERIVED
Land use	Residential — multi-unit	USER_SET
North edge (80 m)	ROAD, collector	USER_SET
Other three edges	ADJACENT_PLOT	USER_SET
FAR maximum	5.0	USER_SET (from plot data sheet)
Height maximum	100.0 m	USER_SET (from plot data sheet)
Coverage maximum	60%	USER_SET (from plot data sheet)
Tower plate cap	40% of plot = 1,280.0 m ²	USER_SET
Road-edge setback	6.0 m	DERIVED — rule R-SETBACK-ROAD-RES
Other-edge setback	4.5 m	DERIVED — rule R-SETBACK-SIDE-RES
Parking counts toward FAR	No	USER_SET — declared; Q1 is OPEN
Ground floor-to-floor	4.5 m	ASSUMED
Podium floor-to-floor	3.5 m	ASSUMED
Typical floor-to-floor	3.2 m	ASSUMED
Podium levels available	2 (+1 basement)	USER_SET
Bay area factor	32.0 m ² /bay	ASSUMED
Podium usable-for-parking fraction	92%	ASSUMED
Realism discount	1.00	USER_SET (default)

A.2 Envelope

Step	Computation	Result	Binding
Setback-permitted footprint	$(80 - 4.5 - 4.5) \times (40 - 6.0 - 4.5) = 71.0 \times 29.5$	2,094.5 m ²	—
Coverage cap	$3,200 \times 0.60$	1,920.0 m ²	—
Podium footprint	$\min(2,094.5, 1,920.0)$	1,920.0 m²	Coverage
Tower plate (selected)	$\leq \min(\text{podium footprint}, 1,280.0)$	1,060.0 m ²	Within cap
Base height	$4.5 + 2 \times 3.5$	11.5 m	—

Step	Computation	Result	Binding
Height available for tower	100.0 – 11.5	88.5 m	Height ceiling

A.3 Capacity bands

Band	Derivation	GFA
A — Regulation-limited	FAR 5.0 × 3,200 = 16,000 permitted. Ground 1,920 + 13 tower floors × 1,060 = 15,700 (a 14th floor would exceed 16,000)	15,700 m²
B — Geometry-limited	[88.5 / 3.2] = 27 tower floors. 1,920 + 27 × 1,060	30,540 m²
C — Parking-limited	Supply 165 bays ÷ 1.519 bays/unit = 108 units → 13.5 floors → 13 floors	15,700 m²
Governing	min(A, B, C)	15,700 m² — Band A (FAR)
Integer granularity loss	16,000 – 15,700	300 m ² (1.88% of permitted)

Headroom: 46.9 m of unused height (14 further floors) exists geometrically but is forbidden by FAR. Parking headroom is 7 bays (4.4%) — near-binding.

A.4 Configuration

Item	Value
Tower floors	13
Total height	11.5 + 13 × 3.2 = 53.1 m (≤ 100.0 ✓)
Units per floor	8 — 4 × 1-bed, 3 × 2-bed, 1 × 3-bed
Unit NSA	1-bed 70, 2-bed 110, 3-bed 160 m ²
NSA per floor	4×70 + 3×110 + 1×160 = 770 m²
Tower plate gross	1,060 m ² (core + circulation + services = 290 m ²)
Tower efficiency	770 / 1,060 = 72.64%
Total units	13 × 8 = 104
Achieved mix	52 / 39 / 13 = 50.0% / 37.5% / 12.5%
Total NSA	13 × 770 = 10,010 m²
Whole-building efficiency	10,010 / 15,700 = 63.76%
Total GFA	1,920 + 13,780 = 15,700 m²
Achieved FAR	15,700 / 3,200 = 4.906 (≤ 5.0 ✓)
Coverage	1,920 / 3,200 = 60.0% (= cap ✓)

Note the two efficiency figures. Tower efficiency 72.64%; whole-building efficiency 63.76%. They differ because the 1,920 m² ground floor carries no saleable area. **This is exactly why FR-DEF-001 exists.** A report quoting "efficiency" without a scope is meaningless, and the gap between these two numbers is 8.9 percentage points — larger than most of the differences developers argue about.

A.5 Parking

Item	Computation	Result
1-bed @ 1.0 bay	52×1.0	52.0
2-bed @ 1.5 bay	39×1.5	58.5
3-bed @ 2.0 bay	13×2.0	26.0
Resident subtotal	[136.5]	137
Visitor @ 15%	$[0.15 \times 137]$	21
Required		158 bays
Required area	158×32.0	5,056 m ²
Supply area	$(1 \text{ basement} + 2 \text{ podium}) \times 1,920 \times 0.92$	5,299 m ²
Supply	$[5,299 / 32.0]$	165 bays
Headroom	$165 - 158$	7 bays (4.4%)

A.6 Assumption register (top 3 by sensitivity)

Rank	Assumption	Value	Class	Impact if perturbed
1	Bay area factor	32.0 m ² /bay	ASSUMED	At 35.0: supply falls to 151 bays < 158 required. Scheme infeasible. Must drop to 12 floors → 96 units, 145 bays required, 14,640 m² GFA. Cost: –1,060 m² GFA, –8 units.
2	Typical floor-to-floor	3.2 m	ASSUMED	Height is not binding (53.1 of 100 m used), so no effect on this plot. Would bind on a plot with a lower height ceiling.
3	Podium usable fraction	92%	ASSUMED	At 88%: supply 158 bays. Headroom falls to 0. Any demand increase makes the scheme infeasible.

The assumption ranked first changes the answer. A 3 m²/bay difference — well within normal variation between basement and podium layouts — removes a floor. This is the register earning its place: without it, the 13-floor answer looks like a fact.

A.7 Invariant results

ID	Check	Result
INV-01	$\Sigma \text{ level GFA} = \text{total GFA}$	$1,920 + 13,780 = 15,700 \checkmark$
INV-02	$\Sigma \text{ unit NSA} + \text{core/circ} = \text{plate}$	$770 + 290 = 1,060 \checkmark$
INV-03	$\text{GFA} / \text{plot} = \text{FAR}$	$15,700 / 3,200 = 4.906 \checkmark$
INV-04	$\Sigma \text{ units by type} = \text{unit count}$	$52 + 39 + 13 = 104 \checkmark$
INV-05	$\Sigma \text{ mix shares} = 1.000$	$0.500 + 0.375 + 0.125 = 1.000 \checkmark$
INV-06	$\text{count} \times \text{mean NSA} = \text{total NSA}$	$104 \times 96.25 = 10,010 \checkmark$
INV-07	$\text{Efficiency} = \text{NSA/GFA at stated scope}$	tower 0.7264, whole 0.6376 $\checkmark$ (both scoped)
INV-08	$\text{Podium} \leq \text{setback-permitted}$	$1,920 \leq 2,094.5 \checkmark$
INV-09	$\text{Podium} \leq \text{coverage cap}$	$1,920 \leq 1,920 \checkmark$
INV-10	$\text{Plate} \leq \text{podium and} \leq \text{cap}$	$1,060 \leq 1,920, \leq 1,280 \checkmark$
INV-11	$\Sigma \text{ heights} = \text{total} \leq \text{ceiling}$	$53.1 \leq 100.0 \checkmark$

ID	Check	Result
INV-12	Bays × factor ≤ available area	5,056 ≤ 5,299 ✓
INV-13	Bays ≥ demand	165 ≥ 158 ✓
INV-16	Governing = min(A,B,C), labelled	min(15,700; 30,540; 15,700) = 15,700, Band A ✓
INV-17	Achieved FAR ≤ permitted	4.906 ≤ 5.0 ✓
INV-18	Composition matches FAR treatment	Parking (5,299 m²) excluded per declaration ✓

All checks pass.

A.8 Five-way claim statement for this example

SELF-CONSISTENCY	PASS – all 18 invariants, all encoded constraints
RULE COVERAGE	40 of 61 identified applicable requirements encoded. 21 deferred, listed in the report. Coverage 66%.
GEOMETRIC VALIDITY	PASS – areas independently recomputed, max deviation 0.00%
REGULATORY VALIDITY	NOT ASSESSED
PROFESSIONAL AGREEMENT	Engine within the two-architect band on 8 of 10 golden plots. Measured inter-architect variance on this community: <to be populated in Phase 0>

30. WHAT WE KNOW

Statements we can support with a citation, a measurement, or direct verification. Nothing in this list is inferred.

30.1 Regulatory and jurisdictional

1. Dubai Law No. (16) of 2023 assigns Dubai Municipality exclusive responsibility for regulating urban planning in the emirate. Free zone and special development zone authorities may impose **additional** requirements on top of DM requirements, and route permit applications through DM for structure- and framework-plan compliance. (*Jurisdiction is overlay composition, not rulebook switching.*)
2. The Dubai 2040 Urban Master Plan is the statutory structure plan under that law.
3. Dubai Law No. (3) of 2026 concerning quality and safety of buildings applies emirate-wide including private development zones and free zones such as DIFC, assigns enforcement to Dubai Municipality, and provides for a central building database.
4. Formal mechanisms exist for exemption from planning and construction regulations, and for modification of permitted land use. (*This is why the variance model exists.*)
5. The affection plan is a high-level general site plan carrying plot number, land-use classification and particular zoning requirements, and indicating which further permissions are required. **It is not guaranteed to carry FAR, height limits, or full setback schedules.**
6. A consolidated Dubai Building Code exists and is issued by Dubai Municipality.

30.2 Data availability

1. Dubai Pulse publishes municipal datasets including building permits, building summary information, building floor level information, building usages, and project/project-application records, with CSV download and authenticated API access under the emirate's data dissemination framework.
2. DLD land and transaction datasets are published through the same channel.

30.3 Verified by our own computation

1. The worked example in Appendix A is arithmetically self-consistent and passes all 18 invariant checks. (*Verified in CI.*)
2. The V1 worked examples were **not** self-consistent — errors of $\sim 2.8\times$ on GFA and $\sim 3\times$ on unit count, with the document's own contradiction checker disagreeing with its own output by $3\times$. (*Verified; those examples are withdrawn.*)
3. Nested evolutionary-over-solver optimization at population 100×50 generations $\times 2$ s inner solve requires ~ 2.8 CPU-hours per run. (*Direct arithmetic.*)

30.4 Technical facts

1. Constraint checking and invariant checking answer different questions. Constraints ask what is permitted; invariants ask whether the arithmetic closes. Only the second catches an impossible configuration.
2. Topological rules (travel distance, exit separation) have no constructive inverse and cannot generate geometry.
3. Implementation independence between validator and generator does not confer semantic independence — both evaluate the same encoded constraint set.

4. Backtest coverage without a sharpness constraint is gameable by widening the interval.
 5. Provenance must be emitted at every layer boundary; retrofitting it requires rewriting every layer.
-

31. WHAT WE ARE ASSUMING

Unvalidated beliefs. Each carries a test and a consequence. **None of these is treated as a fact anywhere in this document, and none may be quoted as one.**

31.1 Dangerous — test in Phase 0

ID	Assumption	Why dangerous	Test	If false
A1	Design-partner land packs contain FAR, height and setbacks	The entire input model depends on it	Phase 0 §23.3 — count of 10	Product's first job is data assembly, not computation. Different product, different plan.
A2	Two qualified architects agree within ~5% on a buildable envelope	Sets the accuracy target and the marketing claim	Week 1, three plots, three architects	Reposition from "accurate" to "consistent and auditable" before any marketing
A3	≥50% of governing residential-tower rules are mechanizable	Sizes the whole business	First 40 rules	Narrow the parameter set; reduce claimed coverage publicly
A4	Rule authoring reaches 8+/day after the schema stabilizes	3× cost swing	Measured throughput by complexity class	Budget 14–18 architect-weeks per 300 rules; raise on that basis
A5	Parking has a determinable, citable treatment under FAR	15–35% capacity swing	Q1, week 2	Stays OPEN REGULATORY QUESTION permanently; user declares per project
A6	Users will engage with provenance rather than treating output as an oracle	The trust thesis	Session observation	Re-examine everything downstream of §13

31.2 Significant — test in Phases 1-2

ID	Assumption	Test	If false
A7	Developers will pay for capacity computation alone	Phase 0 design-partner conversion	Accelerate Phase 1; Phase 0 becomes an internal foundation rather than a product
A8	Exhaustive enumeration over 4–6 typologies is sufficient at production granularity	Phase 1 measurement	Phase 5 becomes necessary; budget \$8.4 mitigations
A9	2–3 developer partners will contribute historical portfolios under contract	Phase 2 partnership effort	Tier B unavailable; historically-observed capacity does not ship; USER_SET discount persists
A10	Public municipal data lacks NSA, mix and permitted FAR	Phase 2 profiling	Good news — Path A becomes viable for calibration and Phase 2 accelerates
A11	Curated parametric typologies capture the majority of viable residential-tower configurations	Architect review, Phase 1	Typology library must expand substantially, or free-form generation becomes necessary far earlier than planned
A12	Standards hard/soft classification can be elicited in a single onboarding session	Phase 1 onboarding	Onboarding cost rises materially; may need to be billable

31.3 Structural — carried, monitored

ID	Assumption	Monitoring
A13	Dubai's regulatory framework is stable enough that a rule base retains value between updates	Q12 — measure actual change frequency

ID	Assumption	Monitoring
A14	A qualified regulatory architect is hireable as a full-time founding-team member	Hiring pipeline; this is the schedule's binding constraint (R-13)
A15	Professional indemnity insurance is obtainable for this product description at a viable premium	Q11, Phase 0 week 8
A16	Approval-outcome data will eventually be capturable from customers, making it the durable moat	Instrument from Phase 1 even though nothing uses it until much later
A17	Non-convex plot geometry can be addressed in Phase 4 without invalidating the Phase 0 kernel	Design review before Phase 4
A18	Architects will act as a distribution channel rather than resisting the product	Phase 1 shared-link engagement

31.4 The assumptions V1 made silently, now surfaced

V1 silent assumption	Status in V2
The affection plan carries development parameters	Rejected. Known false (§30.1 item 5). CRITICAL_INPUT_GAP handling.
Municipal data will supply efficiency and permitted FAR	Rejected. Expected false (A10). Path B is the primary Tier B source.
A "realistic" capacity band could be produced at MVP	Rejected. Removed by schema until §5.4 gates are met.
Every rule can be made constructive	Rejected. Three rule classes (§11.3).
"Most restrictive wins" is universally defined	Rejected. Undefined for enums and topological rules (§11.6).
The validator agreeing with the generator means compliance	Rejected. Five-way claim statement (§16.5).
MVP in 6 months	Rejected. Phase 0 in 8–12 weeks; full vision 18–24 months.

32. Phase 0 Implementation Plan (8-12 weeks)

32.1 Team — four people

Role	FTE	Notes
Regulatory Architect (licensed, Dubai practice)	1.0	Owns the rule base and the definitions annex. The single most important hire. Full team member with veto over rule publication, not an advisor.
Senior Backend / Geometry Engineer	1.0	Geometry kernel, capacity engine, provenance
Full-stack Engineer	1.0	API, frontend, report generation. Owns the invariant and validation modules — different owner from the capacity engine (Principle 4).
Product / Tech Lead	1.0	Scope discipline, data wrangling, design-partner relationship, measurement

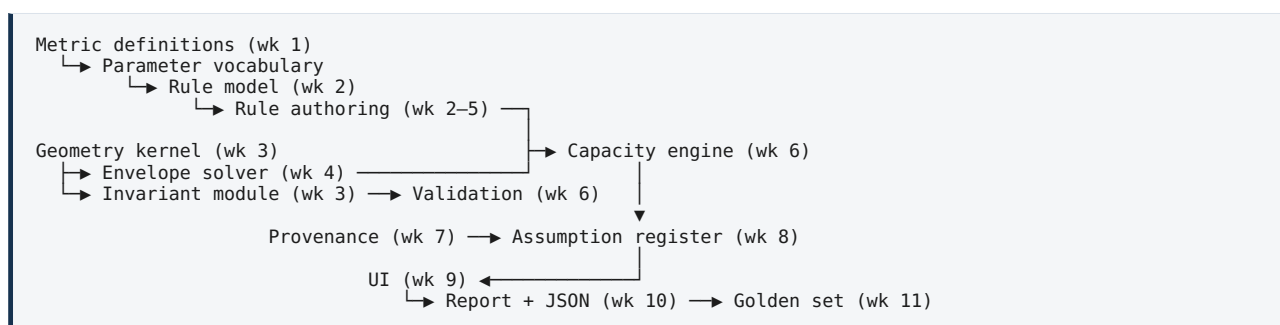
No AI/ML engineer. There is no LLM in Phase 0. **Contracted:** 3 architects for the variance measurement and golden set (~10 days total).

32.2 Schedule

Week	Work	Deliverable
1	Metric definitions annex drafted and signed. Parameter vocabulary for ~40 rules. Inter-architect variance measurement, 3 plots, 3 architects. Secure 10 land packs. Q1 (parking-in-FAR) research begins.	Signed annex; variance figure ; land-pack completeness count
2	Rule record schema, evaluator registry (≤ 12), bitemporal store, test harness, approval workflow. Architect begins authoring. Repo skeleton with import linter.	Rule model + first 10 rules
3	Geometry kernel: per-edge offsets, intersection, area with independent recomputation, validity, exact predicates. Invariant module built alongside, by the other engineer.	Kernel + invariant catalogue
4	Envelope solver: setbacks → footprint → coverage → plate cap → height. Binding-constraint attribution. Rules 10-25.	Envelope solver
5	Parking model: demand → area → levels → headroom → supportable unit ceiling. Rules 25-40.	Parking model; 40 rules complete
6	Capacity bands A/B/C. Governing band. Integer granularity. User realism discount. Independent validation module.	Capacity engine
7	Provenance graph emitted at every boundary. Class assignment. Derivation query.	Provenance
8	Assumption register with perturbation sensitivity. Five-way claim statement. Deferred-check list.	Assumption + claims
9	Frontend: plot form, boundary draw, edge classification, results, click-through provenance tree , assumption register.	UI
10	PDF report + JSON export + round-trip test. CI: invariant checks on all documentation examples.	Outputs
11	Golden-set validation: 10 plots vs. 2 architects each. Diagnose and fix.	Accuracy measurement
12	3 design-partner sessions with observation. Write up the four measurement deliverables.	Findings report

Buffer: weeks 11-12 absorb slip. If rule authoring is slower than assumed (A4), cut to 30 rules rather than extending the schedule — the measurement matters more than the count.

32.3 Critical path



The metric definitions annex is the root. Nothing computes an area term until it is signed. This is a one-week, four-person-hour document that gates the entire project — and skipping it is precisely how V1's example ended up wrong by 3×.

32.4 What Phase 0 delivers beyond software

Four numbers that size the entire business, each of which has a kill/re-scope threshold attached (\$26):

1. **Inter-architect variance** — sets the accuracy target and the marketing claim
2. **Mechanization rate** — sizes the rule base and the honest coverage claim
3. **Rule authoring throughput by complexity class** — sizes the largest cost in the company
4. **Land-pack completeness rate** — tests the input model

A Phase 0 that ships perfect software and measures none of these has failed.

33. Roadmap V1 / V2 / V3

Release	Phases	Duration	Capability	Commercial
V0	0	8–12 wk	Deterministic capacity engine with provenance	1–2 design partners, free
V1	1	+10–14 wk	Configurations, standards, brief, objectives, comparison	3–5 paying customers
V1.5	3	+10–12 wk	AI document extraction with human review	Onboarding cost halved
V2	2	+12–16 wk	Precedent, similarity, benchmarking, historically-observed capacity (gated)	Upgrade; approval-outcome instrumentation begins
V2.5	4	+16–20 wk	Massing, core, floorplate, typical floor (gated on architect panel)	Premium tier
V3	5	+12–16 wk	Multi-objective optimization, Pareto, weight sensitivity (only if enumeration proves insufficient)	Enterprise
V4	—	—	Second jurisdiction; commercial and mixed-use typologies; IFC export; approval-risk model	Expansion

Note the ordering of Phase 3 before Phase 2. AI extraction is placed earlier than precedent because it reduces onboarding cost immediately and has no external data dependency, whereas Phase 2 is gated on partner portfolio acquisition whose timeline is outside our control. Either order works; this one de-risks the schedule.

The second jurisdiction is a V4 decision. Every jurisdiction is a full rule-base build plus a precedent corpus. Expanding geographically before Dubai is profitable multiplies the largest cost in the business before proving it can be recovered.

34. One-Page Brief for the CEO

Why Phase 0 is not "building a calculator"

The one-sentence version: we are not building a calculator — we are building the machine-readable model of Dubai development regulation that every later stage of the product computes against, and the capacity output is simply the first thing that model is capable of producing.

A calculator takes numbers you already have and arithmetic you already agree on. Phase 0 does neither. It takes a plot and produces the numbers, from a body of regulation that currently exists only as PDFs and in the heads of experienced architects. Before Phase 0, "what can I build here?" is a two-to-six-week consultant exercise costing AED 50,000–250,000 that produces one answer with undocumented assumptions. After Phase 0, it is ten seconds, with every number linked to the clause that produced it.

What we are actually building in Phase 0 is the asset the entire product depends on. Four things, none of which is a calculation:

1. **An executable model of Dubai development regulation** — rules as versioned, cited, human-approved, machine-evaluable records. This is the moat. It takes an architect months to build, it cannot be bought, and every regulatory change we absorb widens the gap against a new entrant starting from zero.
2. **A deterministic geometry engine that can *construct* from rules, not merely check against them.** This is the technical distinction between our product and a compliance checker. A checker turns a setback into a test. We turn it into a generator of buildable volume. Phase 4's massing generation and Phase 5's optimization both run on this engine — they do not replace it.
3. **A provenance system that traces every output to a rule, an assumption, or a person.** This cannot be added later. Provenance must be emitted at every layer, so retrofitting it means rewriting every layer. Building it in Phase 0 is the difference between a product an investment committee can rely on and a black box they cannot.
4. **An invariant layer that makes it structurally impossible to publish an impossible building.** Our own earlier specification contained a worked example claiming a tower with three times more units than its floor area could hold, and nothing caught it. That layer now runs on every artifact we produce, including our own documents.

Why not build the AI generation first, since that is what customers imagine? Because you cannot evaluate whether a generated building is good until you can compute, exactly and defensibly, what volume it is allowed to occupy. You cannot calibrate a precedent-based estimate without a deterministic estimate to calibrate against. You cannot review an AI-extracted rule without a rule model to review it into. Each later stage is defined in terms of the stage below it. Building them out of order means building them twice.

And it sells on its own. A developer deciding whether to bid on land needs a capacity number they can defend to their board on Thursday. Today they have a spreadsheet with a rule-of-thumb efficiency factor and no citations. Phase 0 gives them a number with a source next to it, four bands showing which constraint actually binds, and a register of every assumption ranked by how much it moves the answer. No developer in Dubai has that today. It is a product, not a milestone.

What Phase 0 also buys, which is arguably worth more than the software: four measurements that determine the size and shape of everything after it — how much two qualified architects disagree with each other (which tells us whether our claim should be "accurate" or "consistent and auditable"), what fraction of the regulation can actually be mechanized (which sizes the whole business), how fast an architect can author rules (the largest cost we will carry), and

whether the land packs developers actually hold contain the data we need at all. We can commit to the eighteen-month plan honestly only after we have those four numbers. Twelve weeks to de-risk two years is the cheapest option available to us.

The vision has not changed. Plot → regulations → standards → brief → precedent → generate → optimize → validate → outputs. Phase 0 builds the bottom three layers of that stack, properly, once. Everything above it is faster and safer because of it.